Intelligent Multimodal Healthcare Framework for Diabetic Retinopathy Detection, Mental Health Support, and Anti-Cancer Molecule Prediction

A PROJECT REPORT

Submitted by

Deepak Devakumar S [RA2211047010076]
Arvin A[RA2211026010529]
Suriya M[RA2211047010098]

Under the Guidance of

Mrs. Ezhilarasi J
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DEPARTMENT OF COMPUTATIONAL INTELLIGENCE

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Student Name	**: Deepak Devakumar S, Suriya M, Arvin A**
Registration Number	**: RA2211047010076, RA2211047010098, RA2211026010529**
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SUPERVISOR ASSISTANT PROFESSOR DEPARTMENT OF COMPUTATIONAL INTELLIGENCE	**PROFESSOR & HEAD** DEPARTMENT OF COMPUTATIONAL INTELLIGENCE
Examiner 1	Examiner 2

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Deepak Devakumar S
Arvin A
Suriya M

ABSTRACT

There is an increasing demand for accessible health care services and accelerated drug discovery, and a need for intelligent systems that can bring together multiple biomedical data streams. This work provides an integrated multimodal AI service that brings together predictive healthcare and anti-cancer molecule discovery services. It uses machine learning, deep learning and explainable AI to provide clinical and molecular predictions. Our platform's medical service is a combination of image diagnosis, conversational AI, and service integration. This early detects diabetic retinopathy from retinal fundus images using a convolutional neural network and illustrates the results with visual explanations. It also has an AI-powered conversational agent for real-time psychological help via natural language processing and speech recognition. The system also has a location-based doctor finder to find doctors nearby and receive timely care. This bridges the gap between an online diagnosis and an offline treatment. The anti-cancer module also has descriptors and feature engineering to predict the behaviour of small molecules. Ensembling approaches are used to train and merge several machine learning and deep learning models to improve the accuracy of the predictions. It then presents these results in a web tool that allows users to view and make decisions immediately. The proposed system demonstrates a scalable and transparent way of using multimodal data sources in today's healthcare and drug discovery. These include medical images, molecular features and user interactions. This is a system that demonstrates the potential of AI to provide early diagnoses, assistance and data-driven biomedical insights.

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ABBREVIATIONS

AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
CNN	Convolutional Neural Network
MLP	Multilayer Perceptron
XGBoost	Extreme Gradient Boosting
LightGBM	Light Gradient Boosting Machine
SHAP	SHapley Additive explanations
ROC-AUC	Receiver Operating Characteristic – Area Under Curve
PR-AUC	Precision-Recall Area Under Curve
IC50	Half Maximal Inhibitory Concentration
pIC50	Negative Logarithm of IC50
MI	Mutual Information
PCA	Principal Component Analysis
SVM	Support Vector Machine
OOF	Out-of-Fold
NLP	Natural Language Processing
SMILES	Simplified Molecular Input Line Entry System
FDA	Food and Drug Administration
CV	Cross Validation

CHAPTER 1
INTRODUCTION

1.1 Introduction

Artificial Intelligence (AI) has emerged as one of the most significant technological advancements in healthcare and biomedical research. The integration of Machine Learning (ML) and Deep Learning (DL) techniques has enabled the development of intelligent systems capable of analysing complex medical and molecular data with high efficiency and accuracy. These technologies have shown promising applications in disease diagnosis, healthcare assistance, patient engagement, and drug discovery. However, many existing systems are designed to address only specific healthcare tasks, such as clinical diagnosis, conversational support, or molecular activity prediction, without providing a unified and integrated framework.

Healthcare systems continue to face several critical challenges. Early diagnosis of diseases such as diabetic retinopathy is essential to prevent severe complications, yet access to medical experts and diagnostic facilities remains limited, particularly in rural and low-resource regions. In addition, mental healthcare services are often underutilized due to social stigma, lack of awareness, and restricted accessibility. Although AI-based chatbots have been introduced to provide mental health support, many existing solutions lack contextual understanding and proper integration with healthcare services. These limitations highlight the growing need for intelligent healthcare systems capable of combining diagnostic support, user interaction, and healthcare connectivity within a single platform.

At the same time, drug discovery remains a complex and resource-intensive process, especially in the field of anti-cancer research. The identification of biologically active anti-cancer compounds requires the analysis of large molecular datasets, extraction of significant chemical features, and accurate prediction of compound activity. Traditional experimental methods are expensive and time-consuming, increasing the importance of computational approaches in accelerating the discovery process. Machine learning and deep learning techniques provide scalable solutions for molecular screening and prediction, but they require effective feature engineering, robust predictive models, and interpretability to ensure reliable and meaningful outcomes.

To address these challenges, this project proposes an integrated multimodal AI framework that combines predictive healthcare services and anti-cancer molecule discovery within a unified system. The framework is designed to process multiple forms of data, including medical images, molecular descriptors, and user interactions, thereby enabling comprehensive healthcare intelligence. The healthcare component, implemented through the GenNeuro system, integrates medical image analysis, conversational AI, and healthcare assistance services. A Convolutional Neural Network (CNN) is employed to analyze retinal fundus images for the early detection of diabetic retinopathy, providing rapid and non-invasive diagnostic support. In addition, the system incorporates MindMate, an AI-powered conversational agent that utilizes Natural Language Processing (NLP) and speech recognition to provide real-time mental health assistance. A geo-location-based doctor recommendation module is also integrated to help users locate nearby healthcare professionals, effectively bridging the gap between online diagnosis and offline medical consultation.

The anti-cancer prediction component focuses on predicting the biological activity of small molecules using structured molecular features. Various feature engineering techniques, including descriptor-based representations, Mutual Information-based feature selection, model-based feature selection, and dimensionality reduction methods, are employed to improve data representation and model performance. These features are utilized by multiple machine learning and deep learning models, including ensemble learning strategies and neural networks, to enhance prediction accuracy and generalization capability. Ensemble-based approaches improve model robustness by combining the strengths of different predictive techniques.

Furthermore, the proposed framework emphasizes transparency and interpretability through the integration of Explainable AI (XAI) techniques such as feature importance analysis, attention maps, and heatmap visualizations. These methods improve the reliability and trustworthiness of the system by providing insights into model predictions. Interactive web-based interfaces are also developed for both healthcare and anti-cancer modules to enable real-time prediction, visualization, and user interaction, thereby improving the practical accessibility of advanced AI technologies.

In conclusion, this work presents a scalable and comprehensive AI-driven framework that integrates disease diagnosis, mental health support, and anti-cancer drug discovery within a single platform. By combining multimodal data processing, explainable AI methodologies, and full-stack system development, the proposed framework contributes toward the advancement of intelligent, integrated, and user-centered healthcare technologies.

1.2 Problem Statement

The rapid advancement of Artificial Intelligence (AI) in healthcare and biomedical research has led to the development of specialized systems for disease diagnosis, patient engagement, and drug discovery. However, most of these systems are designed independently to address specific problems within individual domains, resulting in limited integration between healthcare services and biomedical research applications. This isolated approach restricts the practical impact of AI systems, as real-world healthcare challenges often require the combined analysis of clinical diagnosis, patient interaction, and molecular-level prediction.

One of the major challenges in healthcare is the early detection of diseases such as diabetic retinopathy, particularly in developing and low-resource regions. Existing diagnostic methods rely heavily on manual examination by medical experts, making the process time-consuming, expensive, and difficult to access for large populations. Although deep learning approaches have demonstrated strong performance in medical image analysis, many existing systems lack interpretability, real-time accessibility, and integration with patient-oriented healthcare services. Furthermore, most diagnostic systems do not provide connectivity with practical healthcare support mechanisms, such as identifying nearby doctors for consultation and follow-up care.

Another significant issue exists in the field of mental healthcare. Despite increasing awareness regarding mental health and emotional well-being, access to timely and personalized mental healthcare support remains limited. Current conversational AI systems often provide generalized responses without understanding emotional, situational, or contextual information. In addition, these systems are usually not integrated with broader healthcare services, limiting their effectiveness in delivering meaningful and continuous support to users.

In biomedical research, the identification of anti-cancer molecules remains a complex, time-consuming, and expensive process. Traditional laboratory-based methods for discovering biologically active compounds require extensive experimentation and resources, making large-scale screening impractical. Machine learning (ML) and deep learning (DL) techniques have emerged as efficient computational alternatives for molecular activity prediction. However, several challenges remain unresolved, including effective molecular feature representation, inconsistent model performance across datasets, poor generalization capability of individual models, and limited use of ensemble learning approaches. Moreover, many existing prediction systems lack interpretability, reducing the reliability and transparency of molecular activity predictions.

A further limitation is the absence of integrated AI systems that combine healthcare diagnosis, conversational assistance, and molecular prediction within a unified framework. Existing approaches generally focus on individual tasks and fail to integrate multimodal data sources such as medical images, user interactions, and molecular descriptors. This lack of multimodal integration limits the development of comprehensive AI-driven healthcare solutions capable of addressing both patient-level and research-level challenges simultaneously.

Additionally, many AI-based healthcare and biomedical research systems remain confined to experimental and research environments without practical deployment for real-world use. Limited user interaction, lack of real-time prediction capability, poor scalability, and the absence of user-friendly interfaces reduce their accessibility and usability. Therefore, there is a strong need for intelligent systems that not only provide high predictive performance but also ensure accessibility, interpretability, scalability, and real-time interaction.

To address these challenges, this project proposes the development of a multimodal AI framework that integrates disease diagnosis, mental healthcare assistance, and anti-cancer molecule prediction into a single unified platform. The proposed system aims to process multiple forms of data, provide interpretable predictions, support real-time interaction, and deliver practical healthcare and biomedical research assistance through an accessible and user-friendly interface.

1.3 Motivation

The primary motivation behind this project is the increasing need for intelligent, accessible, and integrated AI-driven solutions in healthcare and biomedical research. Existing AI systems are typically designed as standalone applications, where one system performs disease diagnosis, another provides conversational assistance, and another predicts molecular activity. However, these systems generally operate independently without effective integration, limiting their overall practical impact and usefulness in real-world healthcare environments.

Early disease detection remains one of the most important challenges in modern healthcare. Diseases such as diabetic retinopathy can be effectively treated if identified during the early stages, whereas delayed diagnosis may lead to severe complications, including permanent vision loss. However, access to specialized medical equipment and experienced healthcare professionals is often limited, especially in remote and underdeveloped regions. This creates a need for intelligent diagnostic systems capable of assisting users through automated disease detection while also guiding them toward appropriate medical consultation and healthcare support.

Mental healthcare accessibility is another major motivating factor for this work. Many individuals experiencing mental health challenges are unable or unwilling to access professional therapy due to social stigma, financial limitations, or lack of available healthcare services. Although conversational AI technologies have been introduced to provide mental health support, many existing systems rely on scripted interactions and lack contextual understanding of human emotions and conversations. More advanced and context-aware conversational systems are therefore required to provide meaningful support and improve user engagement.

In the field of biomedical research, the process of discovering new anti-cancer molecules is highly complex, expensive, and time-intensive. Conventional experimental approaches can analyse only a limited number of molecular compounds and require extensive laboratory resources. Machine learning and deep learning techniques provide the capability to analyse large molecular datasets, identify hidden patterns, and accelerate the discovery of biologically active compounds. However, achieving accurate predictions requires effective feature engineering, robust model development, and the integration of multiple predictive techniques to improve performance and reliability.

Another motivating factor is the gap between AI research and practical implementation. Many advanced AI models developed in research environments are not easily accessible to healthcare professionals, researchers, or patients. The absence of deployable systems, real-time prediction interfaces, and intuitive visualization tools limits the practical adoption of AI technologies. Developing interactive web-based platforms capable of delivering real-time predictions and user-friendly interactions can significantly improve the accessibility and usability of AI-driven healthcare systems.

Furthermore, healthcare and biomedical research involve multiple forms of data, including medical images, molecular fingerprints, and conversational interactions. Integrating these diverse data types within a single intelligent framework can provide a more comprehensive understanding of healthcare challenges and improve decision-making capabilities. Therefore, the motivation of this project is to develop a scalable, interpretable, and deployable multimodal AI system capable of addressing both healthcare and biomedical research challenges in real-world environments.

1.4 Sustainable Development Goal of the Project

This project is aligned with the United Nations Sustainable Development Goals (SDGs) SDG 3: Good Health and Well-being and SDG 9: Industry, Innovation, and Infrastructure. This project leverages Artificial Intelligence to improve health care access and encourage innovation in biomedical research, bringing together technology and meaning to tackle problems in both clinical and research settings.

SDG 3: Good Health and Well-being

SDG 3 seeks good health and well-being for all ages. Our project supports this objective through the development of an integrated AI system that supports early diagnosis, promotes psychological well-being and improves access to health care.

The research solution aims to automatically identify the early stages of diabetic retinopathy by analysing retinal fundus images with deep learning. Early detection helps prevent vision loss and other complications, enhancing patients' quality of life. Further, the platform's conversational AI delivers accessible mental health support to provide emotional advice and mental health resources to patients who may otherwise not have access to psychological support due to cost, stigma or availability.

Finally, a geo-location "Doctor Finder" supports the transition to the physical world by linking patients with local health professionals. This ensures a connection between virtual screening and psychological support, with actual treatment if required. These elements create an infrastructure that increases access to care, promotes early detection and promotes well-being - the core of SDG 3.

SDG 9: Industry, Innovation and Infrastructure

SDG 9 focuses on the need to foster innovation, build robust infrastructure, and encourage sustainable industrialisation. This project will support this SDG by creating a flexible, "scalable" AI system that combines health analytics with computer-driven drug discovery, linking medical practice with biomedical research.

The anti-cancer prediction module employs state-of-the-art machine learning and deep learning techniques for the assessment of molecular descriptors and the prediction of new drug candidates' anti-cancer activity. The system expedites the discovery of promising anti-cancer molecules, saving time and cost in drug discovery with time-consuming and expensive laboratory testing. This approach, which employs ensemble learning and feature engineering, enhances prediction and model performance, enriching biomedical research.

The use of a web-based platform to deliver health and drug discovery tools allows for accessibility, scalability and portability across a range of settings - from clinics to laboratories. The inclusion of multimodal data, explainable AI and interactive real-time capabilities in the system is a smart way to establish sustainable digital infrastructure for the future.

Ultimately, this project contributes to improving public health and building innovation in technology. It promotes SDG 3 through enhanced access to quality healthcare and mental and physical well-being. At the same time, it supports SDG 9 by driving innovation in biomedical AI, building scalable digital infrastructure, and closing the gap between technology, research and practice in healthcare.

CHAPTER 2
LITERATURE SURVEY

2.1 Overview of Research Area

Medical progress has achieved numerous breakthroughs yet cancer still stands as one of the leading global death causes because of its diagnostic and treatment difficulties Murthy and Bethala et al. [36]. The traditional method for discovering cancer drugs requires extensive experimental testing and clinical evaluation processes which develop new cancer treatments at a slow pace and incur high expenses and face challenges from cancer's complex biological makeup Obaido et al. [24]. Drug discovery scientists need to develop better computational methods because traditional techniques fail to effectively investigate chemical compounds that might become new drug treatments Niazi and Mariam et al. [23].

Artificial Intelligence (AI) has undergone significant transformation through Machine Learning (ML) and Deep Learning (DL) advancements which have built new pathways for discovering drugs through computational methods Selvaraj et al. [7]. The methods process extensive molecular databases to uncover the intricate connections that exist between molecular structures and their biological characteristics Mao et al. [10]. The data-driven learning process of ML and DL methods enables researchers to predict anticancer properties of small molecules without conducting time-consuming trial-and-error testing Obaido et al. [24].

The development of small-molecule compounds has become an effective method for fighting cancer because their pharmacokinetic properties enable better treatment results through their ability to enter cells and reach intracellular targets Chandrasekaran et al. [14]. The molecules are represented using molecular descriptors which include two-dimensional (2D) descriptors and molecular fingerprints Yap et al. [15]. The system encodes different structural features of molecules which include physicochemical and topological attributes to enable machine learning algorithms to learn from the data Kaneko et al. [16].

The emergence of biomedical artificial intelligence has created new systems that process multiple data types to enhance prediction accuracy Obaido et al. [24]. The systems use molecular data together with medical imaging data to study disease mechanisms because this approach provides better understanding of complex illnesses Arun et al. [38]. The use of molecular prediction models together with deep learning techniques for image processing through Convolutional Neural Networks (CNNs) enables better predictions of anticancer effects and cancer types which enhances both predictive accuracy and medical application value Li et al. [30].

The research area is moving towards creating advanced AI systems which can scale up their operations while maintaining understandable results through AI systems that process high-dimensional data to produce trustworthy outcomes Cabitza et al. [39]. The development of new anticancer drugs and healthcare decision-making processes has made significant progress through the integration of advanced feature engineering methods together with various machine learning techniques and multimodal learning systems Obaido et al. [24].

2.2 Existing Models and Frameworks

Researchers have dedicated substantial research time during the past few years to develop computational systems which use machine learning and deep learning techniques for discovering novel cancer drugs Obaido et al. [24]. Researchers started their initial research work by using standard machine learning techniques which included Logistic Regression and Support Vector Machines and Decision Trees and K-Nearest Neighbours Shaikhina et al. [35]. The methods used a set of manually developed molecular characteristics in order to forecast biological outcomes with acceptable results when tested on structured datasets Mao et al. [10]. The system struggles to operate effectively because it lacks advanced abilities which would enable it to interpret complex molecular interaction patterns that exist in various chemical datasets with high dimensionality Niazi and Mariam et al. [23].

The latest research employs ensemble learning methods to solve these problems Mohammed and Kora et al. [37]. The ensemble learning methods of Random Forest and Gradient Boosting and XGBoost and LightGBM enable improved prediction ability and system stability Obaido et al. [24]. These models work effectively with high-dimensional descriptor spaces which involve complex interactions among descriptors that appear in structure-activity relationship models Mao et al. [10]. The use of ensemble techniques provides two advantages for computational drug design because they reduce overfitting and improve model accuracy with various data sets Mohammed and Kora et al. [37].

Scientists have developed deep learning as an effective framework to model biological data Li et al. [30]. The three network architectures Multilayer Perceptrons (MLP) and Convolutional Neural Networks (CNN) and Transformers demonstrate exceptional performance in their ability to automatically discover hierarchical patterns Li et al. [30]. CNNs serve as the primary method for applying deep learning to medical imaging data in biomedical research whereas advanced tabular deep learning models such as FT-Transformer and TabNet demonstrate successful performance on tabular molecular data Wang et al. [29]; Shah et al. [27]. These architectures enable learning from both local and global feature interactions, which allows them to predict anticancer properties that traditional methods fail to detect Li et al. [30].

The current models depend on feature selection and dimensionality reduction as their essential element Cai et al. [17]. Scientists need feature selection because molecular descriptors exist at high dimensionality, which makes it necessary for scientists to select particular features that will improve research efficiency while stopping overfitting Cai et al. [17]. The most common methods for feature selection include filter-based methods which use Mutual Information as a selection criterion Hoque et al. [18] embedded methods that utilize L1-regularized models for selection and projection-based methods which include Principal Component Analysis (PCA) as a selection method Omuya et al. [21]. Most studies utilize a single feature selection method, which limits their ability to identify all essential molecular characteristics, but each method offers its unique advantages Haq et al. [19].

Multimodal learning approaches which use data from multiple sources to create better predictions have become a major research focus Obaido et al. [24]. Recent studies in biomedical research demonstrated that when researchers add image features to molecular and clinical data their models achieve better results Arun et al. [38]. Deep learning models can accurately diagnose diseases through their analysis of medical images which include retinal and histopathological images Li et al. [30]. Medical imaging techniques lose important information because they need molecular descriptors to show which data matters most Li et al. [30].

Most existing systems lack a comprehensive framework which unifies various feature representations together with different learning methods and ensemble techniques Mohammed and Kora et al. [37]. The existing models use machine learning and deep learning approaches but they do not take advantage of the combined strengths of both methods Obaido et al. [24]. Researchers have not investigated how to combine multiple feature selection techniques within a single system to achieve better stability and performance across different datasets Haq et al. [19]. The situation calls for an integrated solution which can utilize multiple feature representations together with machine learning and deep learning models and modern ensemble methods to improve cancer prediction results Mohammed and Kora et al. [37].

2.3 Limitations Identified from Literature Survey

Existing studies show substantial advancements achieved through machine learning and deep learning methods which scientists use for developing new cancer treatments Murthy and Bethala et al. [36]. Research face their main obstacle because researchers depend on data from one single source Obaido et al. [24]. Research uses two different methods which focus on either molecular descriptors or medical imaging data but do not combine both approaches Arun et al. [38]. The models face reduced ability to predict results because they cannot assess complete biological information Obaido et al. [24].

The system encounters problems because it cannot handle new data which includes different types of features Niazi and Mariam et al. [23]. Researchers create models which depend on one specific molecular descriptor type, which restricts their ability to function with various data sets Mao et al. [10]. The system develops overfitting problems because it uses multiple data types which people encounter in actual situations Mohammed and Kora et al. [37].

The existing research. The common method uses one technique which involves either correlation filtering or principal component analysis Omuya et al. [21]. The techniques fail to handle non-linear relationships and feature interactions which leads to performance declines Cai et al. [17].

The basic voting and averaging methods represent the most common techniques that researchers use for ensemble learning Mohammed and Kora et al. [37]. The methods fail to achieve better results because they do not utilize the full potential of different machine learning and deep learning systems Mohammed and Kora et al. [37]. Biomedical systems use accuracy as the primary performance measure while giving less consideration to recall and other vital assessment factors Rainio et al. [26]. The approaches require a comprehensive framework which needs to combine multiple data sources together with different feature selection methods and advanced ensemble techniques to improve prediction accuracy and stability and result interpretability Obaido et al. [24].

2.4 Research Objectives

The primary objective of the research study aims to develop a quick and precise computational method which uses advanced machine learning and deep learning technologies to forecast the anticancer effects of small molecules. The research investigates conventional methods that rely on data-driven models which can accurately determine the complex connection between molecular structure and biological activity.

The research study intends to test and analyze different molecular representations which include two-dimensional (2D) descriptors and hybrid descriptors to determine their impact on predictive model performance. The research study intends to enhance feature quality through various feature selection techniques which include Mutual Information, L1-based embedded feature selection, Random Forest ranking, and Principal Component Analysis.

The research aims to evaluate how different machine learning ML and deep learning DL models perform with identical feature sets. The research tests multiple algorithms until it determines which ones most accurately predict anticancer activity. The study creates an all-encompassing ensemble system that combines various machine learning and deep learning methods to enhance prediction accuracy and model generalization abilities. The research establishes performance optimization as the main goal because it helps researchers discover all active anticancer compounds which serve vital functions in biomedical research.

The research establishes interpretability methods as essential tools which provide insights into model predictions while enhancing model understanding. The research project aims to develop a trustworthy system which can grow to support various real-world applications through its ability to integrate new data sources and operate in healthcare facilities and research institutions.

2.5 Product Backlog

Table 2.1 Product Backlog of Anticancer Prediction

S.No	User Stories
#US 1	Filtering molecules using IC50 values
#US 2	Preprocessing the datasets
#US 3	Collecting and analyzing previous research papers related to anticancer prediction
#US 4	Classifying active and inactive anticancer molecules
#US 5	Performing feature selection using ALT400, MI400, and PCA400
#US 6	Training a total of 102 machine learning models
#US 7	Extracting molecular features for each compound

#US 8	Performing 5-fold stratified cross validation
#US 9	Selecting the top 10 machine learning models based on recall and model diversity
#US 10	To evaluate and filter the top machine learning models based on recall and accuracy for selecting high-performing models.
#US 11	To train an improved deep MLP model on both datasets and feature tracks to capture nonlinear relationships.
#US 12	To implement and train FT-Transformer models on both datasets and feature tracks for learning feature dependencies.
#US 13	To develop and train TabNet models to enable interpretable and sequential feature learning.
#US 14	To apply hyperparameter optimization using Optuna for improved deep MLP to enhance performance.
#US 15	To apply hyperparameter optimization using Optuna for FT-Transformer to achieve optimal configurations.
#US 16	To apply hyperparameter optimization using Optuna for TabNet to improve model generalization.
#US 17	To develop a healthcare platform interface so that users can interact with the system and access predictions.
#US 18	To integrate the trained models into the platform so that predictions can be delivered in a user-accessible format.
#US 19	Hyperparameter optimization for ML both random search and grid search
#US 20	Validating all the DL models using testing dataset
#US 21	Performing 8 ensemble techniques for 12 DL models using the saved artifacts of tuning

#US 22	Performing 8 ensemble techniques for top 10 ML models using the saved artifacts of tuning
#US 23	Performing 8 ensemble techniques between ML and DL models which encomposed of another 70+ model
#US 24	Training a CNN model for predicting diabetics in human eye
#US 25	Selecting the top model which encomposed of 2 DL and 1 ML which produced excellent result and testing with external validation dataset which encomposed of authentic drug compounds saving lot of time and funds for wasting money on non active drug compounds

2.6 Plan of Action (Project Roadmap)

This research project aims to build a multimodal application of artificial intelligence for predicting anti-cancer molecules by integrating chemical data analysis with machine learning, deep learning and ensemble methods.The plan details the successive steps in data processing, model building, validation and deployment for a structured approach and replicable results as shown in Table 2.1.

Phase 1: Review and Problem Statement

Duration: Week 1–2

To review the literature on anti-cancer drug discovery involving machine learning, deep learning and multimodal approaches. Examine various molecular descriptors, feature selection approaches and models. Recognize the limitations in absence of multimodal integration, poor generalization and ineffective feature selection methods. Propose a research problem and objectives.

Deliverables:

Literature review, literature gaps identification, finalize research problem and research methodology.

Phase 2: Data Collection and Preprocessing

Duration: Week 3–4

Goal:

To gather molecular data from PubChem with compounds and IC50 values. Transform IC50 to pIC50 and classify compounds as active or inactive. Clean the dataset by removing duplicates, filling missing data and removing irrelevant features. Prepare the data to be clean, consistent and ready for modelling.

Deliverables:

Data cleaning, labelled data (active/inactive) and molecular data preprocessing for feature extraction.

Phase 3: Feature Engineering and Dataset Preparation

Duration: Week 5–6

Goal:

To calculate molecular descriptors such as 2D and hybrid descriptors (2D + fingerprints). Use various feature selection methods including Mutual Information (MI-400), L1-based Logistic Regression with Random Forest ranking (ALT-400) and Principal Component Analysis (PCA-400). Provide various feature sets for comparative study.

Deliverables:

Datasets with feature engineering (MI-400, ALT-400, PCA-400), and formatted datasets for training.

Phase 4: ML Model Development

Duration: Week 7–8

Goal:

To build several machine learning models such as Logistic Regression, Support Vector Machine (SVM), Random Forest, XGBoost, LightGBM and etc. Use stratified cross-validation for evaluation. Evaluate models with different metrics like accuracy, recall, F1, ROC-AUC, and PR-AUC.

Deliverables:

Machine learning models, cross-validation scores, comparison report.

Phase 5: Deep Learning Model Development

Duration: Week 9-10

Goal:

To build deep learning models like Multilayer Perceptron (MLP), FT-Transformer and TabNet for feature interactions. Enhance model learning using techniques like hyperparameter tuning and learning rate scheduling.

Deliverables:

Deep learning models, model optimization and evaluation metrics.

Phase 6: Ensemble Learning and Integration

Duration: Week 11

Goal:

To ensemble machine learning and deep learning models, including voting, weighted averaging and rank aggregation. Enhance prediction accuracy and robustness by combining advantages of models.

Deliverables:

Ensemble models, better prediction performance and analysis of ensemble performance.

Phase 7: Validation and Analysis

Duration: Week 12

Goal:

To cross-validate the final model on external data and evaluate its predictions. Create plots (e.g. performance, feature importance). Use explainability methods, like SHAP, to explain predictions.

Deliverables:

Validation report, performance visuals, explainability results and final report.

Phase 8: System Integration and Deployment

Duration: Week 13

Goal:

To deploy the models in a system that has a front-end and back-end. Allow for live prediction and interaction. Make the system scalable and practical for use.

Deliverables:

System deployment, frontend, backend integration, and project complete.

CHAPTER 3

SPRINT PLANNING AND EXECUTION

3.1 SPRINT I – Dataset Preparation and Baseline Model Development

3.1.1 Objectives with User Stories of Sprint I

Sprint I mainly focused on building the foundation for the anticancer prediction system by preparing datasets, extracting molecular features, performing feature selection, and training baseline machine learning models. The objective of this sprint was to create a clean and reliable data pipeline that could support further machine learning, deep learning, and ensemble experiments in later sprints.

In this sprint, preprocessing techniques were applied to molecular datasets to improve data quality and remove inconsistencies. Feature extraction and feature selection methods were used to identify the most informative molecular descriptors related to anticancer activity. Multiple baseline machine learning models were trained and evaluated to identify strong-performing algorithms for future ensemble learning.

Table 3.1: Detailed User Stories of Sprint I

S.NO	Detailed User Stories
#US 1	Filtering molecules using IC50 values
#US 2	Preprocessing the datasets
#US 3	Collecting and analyzing previous research papers related to anticancer prediction
#US 4	Classifying active and inactive anticancer molecules
#US 5	Performing feature selection using ALT400, MI400, and PCA400
#US 6	Training a total of 102 machine learning models
#US 7	Extracting molecular features for each compound
#US 8	Performing 5-fold stratified cross validation
#US 9	Selecting the top 10 machine learning models based on recall and model diversity

3.1.2 Functional Document

3.1.2.1 Introduction

This sprint focuses on preparing a high-quality molecular dataset and building baseline machine learning models for anticancer activity prediction. The sprint includes dataset preprocessing, feature extraction, feature selection, model training, and performance evaluation.

The main purpose of this sprint is to ensure that the datasets are clean, standardized, and suitable for training reliable predictive models. Different feature selection methods and machine learning algorithms were explored to identify strong baseline models for future optimization and ensemble learning.

3.1.2.2 Product Goal

The goal of this sprint is to develop a strong baseline prediction pipeline for identifying anticancer compounds using molecular descriptors and machine learning techniques as shown in Table 3.1.

The sprint aims to:

- Prepare clean and standardized datasets
- Extract molecular descriptors and fingerprints
- Perform feature selection to reduce dimensionality
- Train and evaluate multiple machine learning models
- Identify top-performing models for further optimization
- Build a reliable foundation for deep learning and ensemble modelling

3.1.2.3 Demography

Users:

- Biomedical researchers working on anticancer drug discovery
- Data scientists working with molecular datasets
- Pharmaceutical researchers analyzing chemical compounds
- Academic researchers and students in AI-driven drug discovery

Location:

- Research laboratories and universities
- Pharmaceutical research centers
- AI and bioinformatics research environments
- Global research environments working on cancer therapeutics

3.1.2.4 Business Processes

Dataset Preparation

- Collect molecular datasets from PubChem BioAssay
- Filter molecules based on IC50 values
- Convert IC50 values into pIC50 values
- Label compounds as active or inactive

Data Preprocessing

- Remove duplicate and invalid entries
- Handle missing values using median imputation
- Remove low-variance features
- Standardize dataset structure

Feature Engineering

- Generate molecular descriptors and fingerprints
- Create 2D and Hybrid datasets

Feature Selection

- Apply MI400 feature selection
- Apply ALT400 feature selection
- Apply PCA400 dimensionality reduction

Model Development

- Train multiple machine learning models
- Perform 5-fold stratified cross-validation
- Evaluate models using performance metrics

Model Selection

- Select top-performing models based on recall and model diversity

3.1.2.5 Features

Feature 1: Molecular Dataset Preparation

Description

This feature prepares molecular datasets by filtering compounds based on IC50 values and converting them into structured datasets for anticancer prediction.

User Story

As a researcher, I want properly filtered molecular datasets so that model training becomes reliable and meaningful.

Feature 2: Data Preprocessing Pipeline

Description

This feature performs preprocessing tasks such as missing value handling, duplicate removal, and variance filtering to improve dataset quality.

User Story

As a data scientist, I want automated preprocessing so that I can reduce errors and improve model performance.

Feature 3: Molecular Feature Extraction

Description

Molecular descriptors and fingerprints are extracted using descriptor-generation tools to represent chemical properties numerically.

User Story

As a researcher, I want molecular features extracted automatically so that models can learn chemical patterns effectively.

Feature 4: Feature Selection

Description

Feature selection methods such as MI400, ALT400, and PCA400 are used to reduce dimensionality and retain important molecular descriptors.

User Story

As a data scientist, I want important features selected automatically so that models can train efficiently and avoid overfitting.

Feature 5: Machine Learning Model Training

Description

A total of 102 machine learning models are trained using different algorithms and feature tracks to identify strong-performing models.

User Story

As a researcher, I want to compare multiple ML algorithms so that I can identify the best-performing models.

Feature 6: Cross Validation and Evaluation

Description

5-fold stratified cross-validation is used to evaluate model stability and performance consistency.

User Story

As a researcher, I want reliable validation methods so that model performance can be trusted on unseen data.

Feature 7: Top Model Selection

Description

Top-performing models are selected mainly based on recall while also maintaining diversity for future ensemble learning.

User Story

As a researcher, I want the best models selected carefully so that future ensembles become more robust and accurate.

3.1.3 Architecture Document

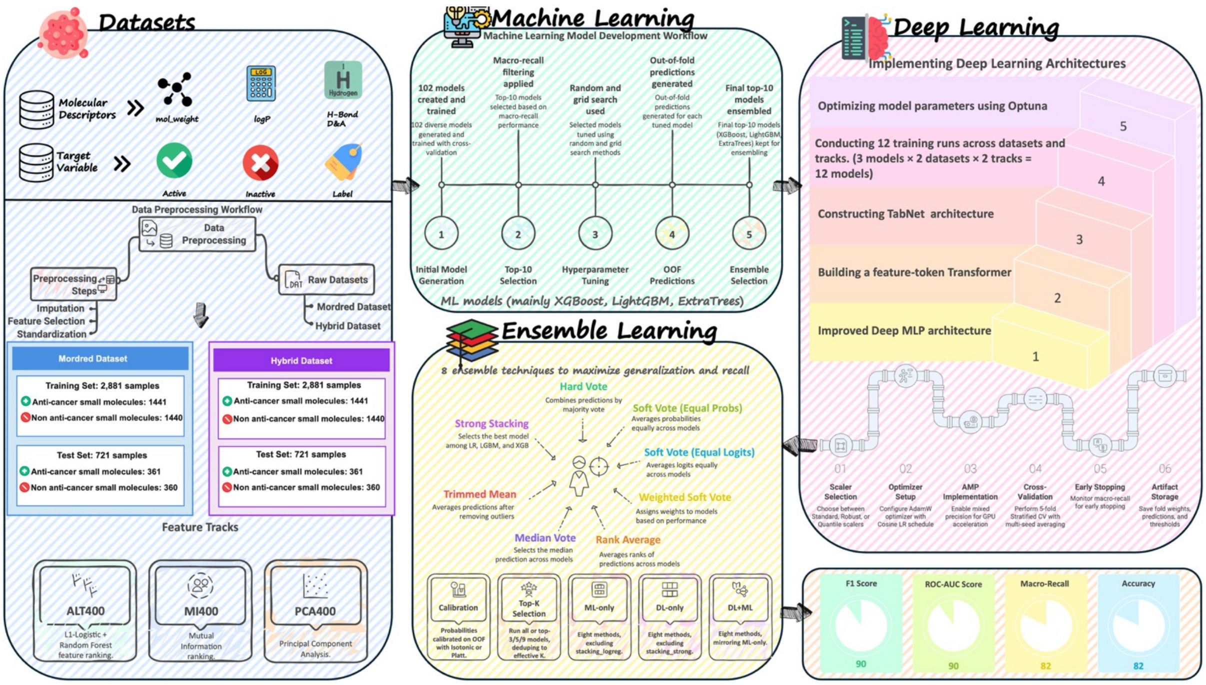


Figure 3.1: Architecture Diagram

3.1.3.1 Application Architecture

The Sprint I architecture was designed to support data preprocessing, feature engineering, feature selection, and baseline machine learning model training as shown in Figure 3.1.

Dataset Processing Layer

- Molecular datasets collected from PubChem BioAssay
- IC50 filtering and pIC50 conversion
- Active and inactive compound labeling

Preprocessing Layer

- Missing value imputation

- Duplicate removal
- Variance threshold filtering

Feature Engineering Layer

- Molecular descriptor extraction
- Fingerprint generation
- Creation of 2D and Hybrid datasets

Feature Selection Layer

- MI400 selection
- ALT400 selection
- PCA400 transformation

Machine Learning Layer

- Training 102 ML models
- Cross-validation evaluation
- Performance metric calculation

Model Selection Layer

- Recall-based ranking
- Selection of top 10 ML models

3.1.3.2 System Architecture

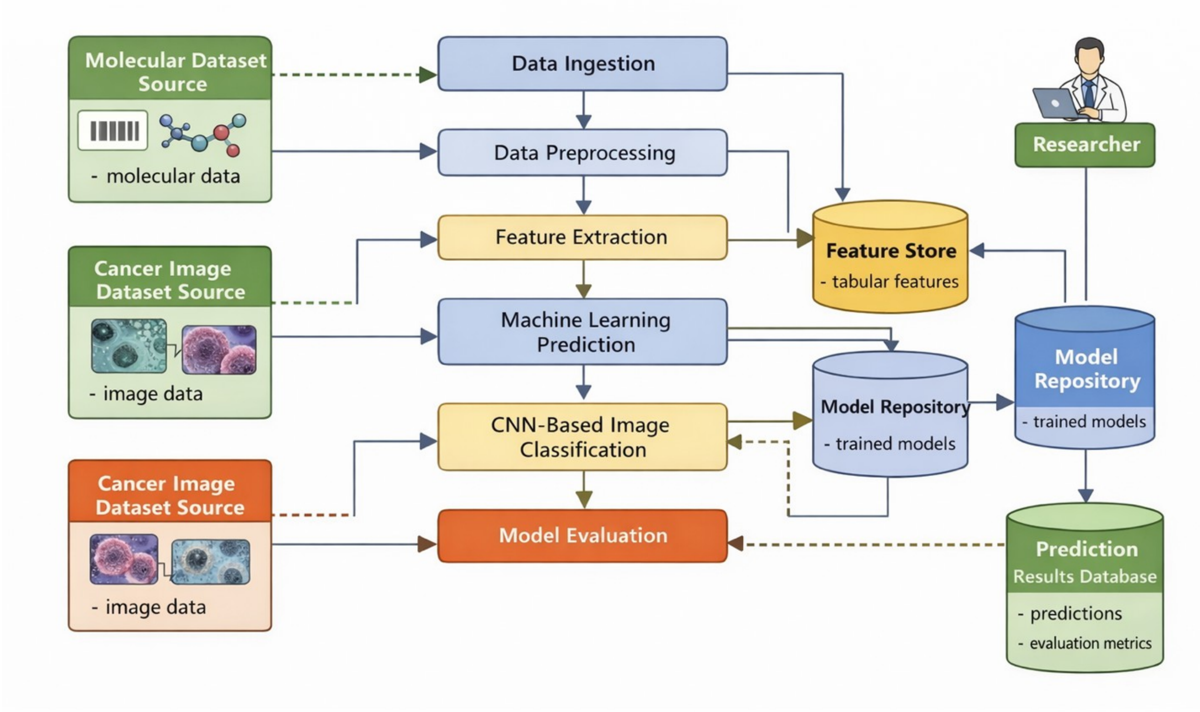


Figure 3.2: Data Flow Diagram

The Sprint I system architecture consists of the following layers as shown in Figure 3.2:

Input Layer

- Molecular datasets
- SMILES representations
- IC50 activity values

Processing Layer

- Data cleaning
- Feature extraction
- Feature preprocessing

Intelligence Layer

- Feature selection algorithms
- Machine learning models

Evaluation Layer

- Cross-validation
- Performance metric computation

Output Layer

- Model evaluation reports
- Performance plots
- Selected top models

3.1.3.3 Data Exchange

Frequency of Data Exchange

- Dataset preprocessing occurs during pipeline execution
- Feature selection outputs are generated after preprocessing
- Model predictions are generated during training and validation

Data Types

Table 3.1: Data Types

Data Type	Description
Molecular Data	Chemical descriptors and fingerprints
Processed Data	Cleaned and standardized datasets
Feature Data	Selected molecular descriptors
Model Outputs	Predictions and evaluation scores

Mode of Exchange

- Python pipelines for preprocessing and training
- CSV-based dataset storage
- Model evaluation outputs stored as reports and plots

3.1.4 Outcome of the Objectives

3.1.4.1 Dataset Preparation

Objective

To prepare clean molecular datasets for anticancer prediction.

Outcome

The datasets were successfully cleaned, filtered, and standardized for downstream modelling tasks.

3.1.4.2 Feature Extraction and Selection

Objective

To extract meaningful molecular descriptors and reduce dimensionality.

Outcome

Feature extraction and feature selection methods successfully identified informative molecular descriptors for model training.

3.1.4.3 Machine Learning Model Training

Objective

To train multiple baseline machine learning models.

Outcome

A total of 102 machine learning models were trained successfully across different feature tracks and datasets.

3.1.4.4 Cross Validation and Model Evaluation

Objective

To evaluate model stability and predictive performance.

Outcome

5-fold stratified cross-validation ensured stable and reliable model evaluation across datasets.

3.1.4.5 Top Model Selection

Objective

To identify the strongest baseline models for future ensemble learning.

Outcome

Top-performing models were selected based on recall performance and ensemble diversity considerations.

3.1.5 Sprint Retrospective

Sprint Retrospective			
What went well	**What went poorly**	**What ideas do you have**	**How should we take action**
*This section highlights the **successes and positive outcomes from the sprint**. It helps the team recognize achievements and identify practices that should be continued.*	*This section identifies the **challenges, roadblocks, or failures encountered during the sprint**. It helps pinpoint areas that need improvement or change.*	*This section is for brainstorming **new approaches, tools, or strategies to enhance the team's efficiency, productivity, or project outcomes.***	*This section outlines specific steps or solutions to address the issues and implement the ideas discussed, ensuring continuous improvement in future sprints.*
Molecular datasets were cleaned and standardized successfully.	Some datasets contained missing and duplicate values.	Automate preprocessing validation checks.	Add validation scripts before saving processed datasets.
Feature selection reduced dimensionality effectively.	Some preprocessing tasks had to be repeated multiple times.	Maintain preprocessing logs for tracking changes.	Save preprocessing outputs separately with version name
102 ML models were trained successfully	Training multiple models increased computation time.	Use automated experiment tracking.	Store model results automatically after training.
Cross-validation improved reliability of evaluation.	Some feature files were overwritten accidentally.	Maintain proper dataset versioning.	Save datasets using separate version folders
Top models were selected clearly using recall-based ranking.	Comparing many models manually was time-consuming	Create automated ranking summaries.	Generate model comparison reports automatically.

Figure 3.3: Sprint Retrospective of Mixed Cropping System

3.2 SPRINT II - Model Development and Optimization

3.2.1 Objective with User Stories of Sprint I

Machine learning models development together with advanced deep learning system creation will lead to better predictive results in anticancer molecule classification. The phase involves finding superior models while multiple deep learning systems undergo training and their performance gets enhanced through automated hyperparameter optimization. The system needs to improve its classification capacity because it needs to use complex nonlinear molecular descriptor relationships which exist in its data.

The sprint begins with the evaluation of previously trained machine learning models, where models are ranked based on key performance metrics such as recall and accuracy.407 The evaluation results show that top-performing models pass through filtering which selects only the most trustworthy models for additional evaluation. The step helps to decrease model duplication while directing computer resources toward evaluating the most promising candidates.

The system moves to its next development stage when developers select deep learning system components which they will create and train as shown in Table 3.3. The advanced Deep Multilayer Perceptron (MLP) model possesses the ability to handle nonlinear feature interactions among different features. The researchers developed advanced architectures which they trained on two datasets and specific feature tracks. The selected models possess the ability to handle structured data while their attention mechanisms enable them to comprehend intricate feature associations.

The training process uses multiple datasets and different feature representations to maintain test equity and achieve complete assessment. The system conducts separate training for each model which enables it to evaluate their distinct performance attributes.

The team uses Optuna framework for conducting hyperparameter optimization to achieve better results. The process includes automated tuning for essential parameters which include learning rate and batch size and number of layers and regularization methods. The team uses optimization techniques on every deep learning model to determine optimal settings for Improved Deep MLP and FT-Transformer and TabNet architectures.

The sprint focuses on creating a collection of highly efficient models which will serve as the main resources for upcoming processes that involve ensemble learning and final prediction. The current phase plays a vital role in enhancing the system's main forecasting capabilities.

Table 3.3 User Stories of sprint 1 for Model Development

S.No	User Stories
#US 10	To evaluate and filter the top machine learning models based on recall and accuracy for selecting high-performing models.
#US 11	To train an improved deep MLP model on both datasets and feature tracks to capture nonlinear relationships.
#US 12	To implement and train FT-Transformer models on both datasets and feature tracks for learning feature dependencies.
#US 13	To develop and train TabNet models to enable interpretable and sequential feature learning.
#US 14	To apply hyperparameter optimization using Optuna for Improved Deep MLP to enhance performance.
#US 15	To apply hyperparameter optimization using Optuna for FT-Transformer to achieve optimal configurations.
#US 16	To apply hyperparameter optimization using Optuna for TabNet to improve model generalization.
#US 17	To develop a healthcare platform interface so that users can interact with the system and access predictions.

#US 18	To integrate the trained models into the platform so that predictions can be delivered in a user-accessible format.

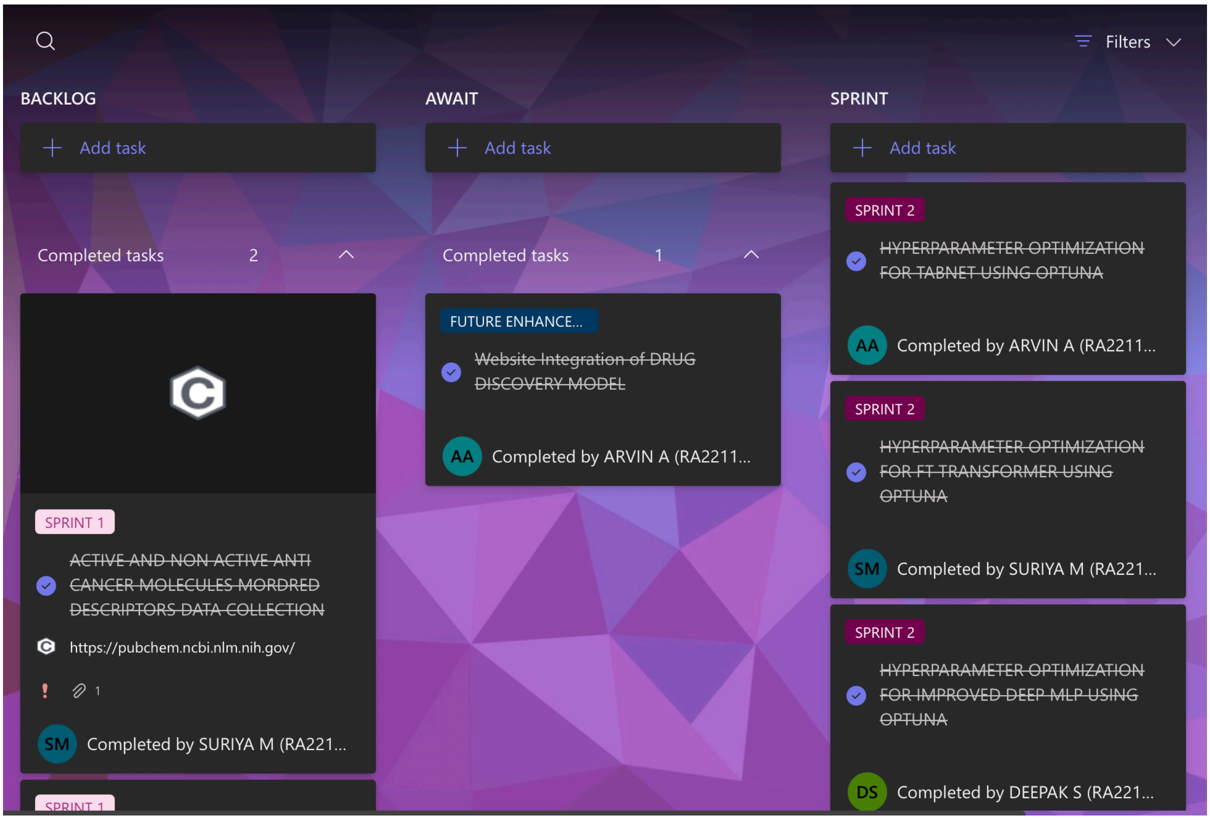


Figure 3.4 User story for sprint 2

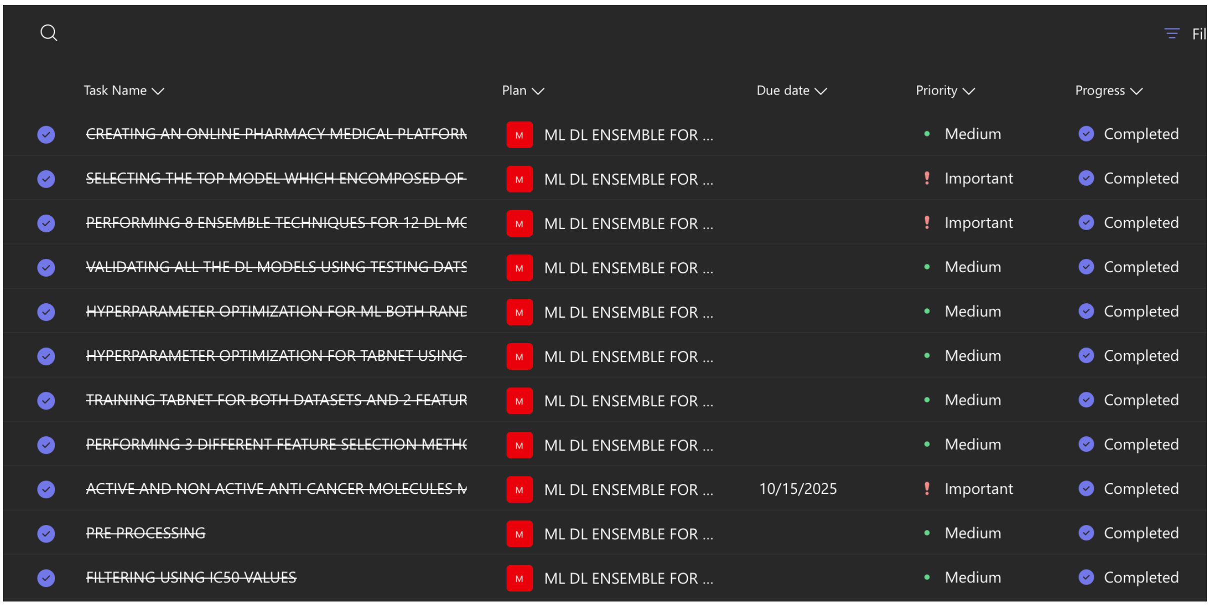


Figure 3.5 User story for Model Development

3.2.2 Functional Document

3.2.2.1. Introduction

The second sprint focuses on enhancing the predictive capabilities of the anticancer molecule classification system through advanced model development and optimization. The classification performance of the system will improve through this phase which implements deep learning architectures and selects advanced machine learning models based on preprocessed datasets and feature-engineered input data.

The system utilizes structured molecular descriptor datasets which are derived from multiple feature selection techniques. The process begins by the team who uses evaluation metrics to choose the best machine learning models through a procedure of filtering and selection. The process ensures that only dependable models get selected for additional research work.

The system uses machine learning models together with deep learning architectures which include an enhanced Deep Multilayer Perceptron system and FT-Transformer system and TabNet system. The models learn from several datasets while they follow different feature tracks to identify complex nonlinear patterns which exist in molecular data. The system uses Optuna for hyperparameter optimization which automatically adjusts model settings to achieve better results. The sprint establishes a fundamental base which enables the creation of accurate predictive models while it readies the system for upcoming ensemble learning and ultimate assessment.

3.2.2.2. Product Goal

Your training provides you with information which extends until October 2023. The sprint aims to enhance the accuracy and robustness and generalization abilities of the anticancer prediction system through the creation and refinement of machine learning and deep learning models. The system employs systematic evaluation methods to discover optimal models which it boosts through automated hyperparameter tuning.

The system is intended to:

- Improve prediction accuracy and recall of anticancer classification
- Capture complex nonlinear relationships in molecular descriptor data
- Provide optimized deep learning models for further ensemble integration
- Ensure consistent performance across multiple datasets and feature tracks

- Support scalable and reproducible model development

3.2.2.3. Demography

Users:

- Researchers who work in drug discovery and computational chemistry
- Data scientists who practice machine learning
- Biomedical researchers who study anticancer treatments
- AI developers who create healthcare solutions

Location:

- Research laboratories and academic institutions
- The pharmaceutical industry and biotechnology sector
- Cross computational research environments
- The system can extend to both worldwide healthcare and research networks

3.2.2.4. Business Processes

The major processes involved in this sprint include:

Model Selection and Filtering:

- Researchers assess various machine learning models through their achieved performance metrics
- Researchers choose the ten best-performing models according to their capacity to recall and their overall accuracy

Deep Learning Model Training:

- Researchers conducted training for Improved Deep MLP model across multiple datasets and different feature tracks
- Researchers conducted training for FT-Transformer model which handles structured data
- Researchers conducted training for TabNet model which allows users to understand how features are learned

Hyperparameter Optimization:

- The team uses Optuna to automate the process of tuning model parameters through its automated tuning system
- Researchers conducted optimization tests to find the ideal learning rate and batch size and architecture depth and regularization settings
- Deep learning models require separate optimization processes for their individual components

Performance Evaluation:

- Researchers evaluate models through accuracy assessment and recall measurement and F1-score calculation and ROC-AUC evaluation and PR-AUC testing
- Researchers found the best-performing configurations through their evaluation process

3.2.2.5. Features

This project focuses on implementing the following key features:

Feature 1: Machine Learning Model Selection

1. Description:
   - The system evaluates multiple machine learning models and selects the top-performing models based on recall and accuracy. This ensures that only reliable and high-quality models are used for further processing.
2. User Story:
   - The researcher intends to identify the best machine learning models so that only high-performing models are used for anticancer prediction.

Feature 2: Improved Deep MLP Model

1. Description:
   - The system implements an enhanced deep Multilayer Perceptron model capable of capturing nonlinear relationships in molecular descriptor data. The model is trained across multiple datasets and feature tracks.
2. User Story:
   - The researcher wants to use a deep learning model that can effectively learn complex patterns in molecular data to improve prediction accuracy.

Feature 3: FT-Transformer Model

1. Description:
   - The system incorporates FT-Transformer architecture to model feature interactions using attention mechanisms. This approach helps in capturing dependencies between molecular descriptors.
2. User Story:
   - The researcher wants to apply transformer-based models to improve learning of feature relationships in structured data.

Feature 4: TabNet Model

1. Description:
   - The system uses TabNet architecture to perform sequential feature selection and provide interpretable results. This model enhances transparency and model performance.
2. User Story:
   - The researcher wants an interpretable deep learning model that can identify important features during prediction.

Feature 5: Hyperparameter Optimization using Optuna

1. Description:
   - The system integrates Optuna for automated hyperparameter tuning of deep learning models. This improves model performance by identifying optimal configurations.
2. User Story:
   - The researcher wants to optimize model parameters automatically to achieve better accuracy and generalization.

Feature 6: Multi-Dataset and Feature Track Training

1. Description:
   - The system trains models across multiple datasets and feature tracks to ensure robust evaluation and comparison.

2. User Story:
    - The researcher wants to train models on different datasets and feature sets to ensure consistent and reliable performance.

3.2.3 Architecture Document

3.2.3.1. Application

Model Architecture:

The Sprint II architecture design implements a microservices framework which allows developers to build scalable systems through modular architectural components which support their deep learning model development work and model selection process and hyperparameter tuning activities within their project work. The system functions through its distinct components which operate as separate services to deliver testing capabilities with flexible system operation and complete system reproduction. The system architecture supports simultaneous execution of multiple models across various datasets and different feature tracks.

The core architecture starts with the Model Selection Service which tests existing machine learning models to select the most effective models through their recall and accuracy performance metrics. The system forwards only dependable high-standard models to the next processing stage.

The most important services included in this architecture are as follows:

- Model Selection Service:

  The service tests various machine learning models through performance assessment which includes recall and accuracy measurements. The system identifies the ten most effective models through its filtering process which keeps only the best models for upcoming evaluations and testing procedures.

- Improved Deep MLP Service:

  The service develops and tests an upgraded deep Multilayer Perceptron model through its implementation across various datasets and feature tracks. The model uses molecular descriptor data to identify nonlinear relationships which enhance its ability to make accurate predictions.

- FT-Transformer Service:

  The service applies a transformer-based architecture to model feature interactions through its attention mechanism usage. The system learns intricate relationships among molecular characteristics which leads to better classification results. • TabNet Service:

The service uses TabNet architecture to execute feature selection in sequence while delivering model results that users can understand. The system improves its efficiency through training which enables it to find essential features that improve both results and visibility.

- Multi-Dataset Training Service:

  This service ensures that all deep learning models are trained consistently across multiple datasets and feature tracks. It enables fair comparison and evaluation of models under different data representations.

- Hyperparameter Optimization Service:

  This service integrates the Optuna framework to perform automated hyperparameter tuning. It explores various parameter combinations such as learning rate, batch size, network depth, and regularization to identify optimal configurations for each deep learning model.

- Model Evaluation Service:

  This service evaluates the performance of both machine learning and deep learning models using metrics such as accuracy, recall, F1-score, ROC-AUC, and PR-AUC. It ensures that models are assessed comprehensively before being used in further stages.

- Model Output Service:

  This service stores the trained and optimized models along with their performance metrics. The outputs generated in this stage are used for subsequent ensemble learning and final prediction processes.

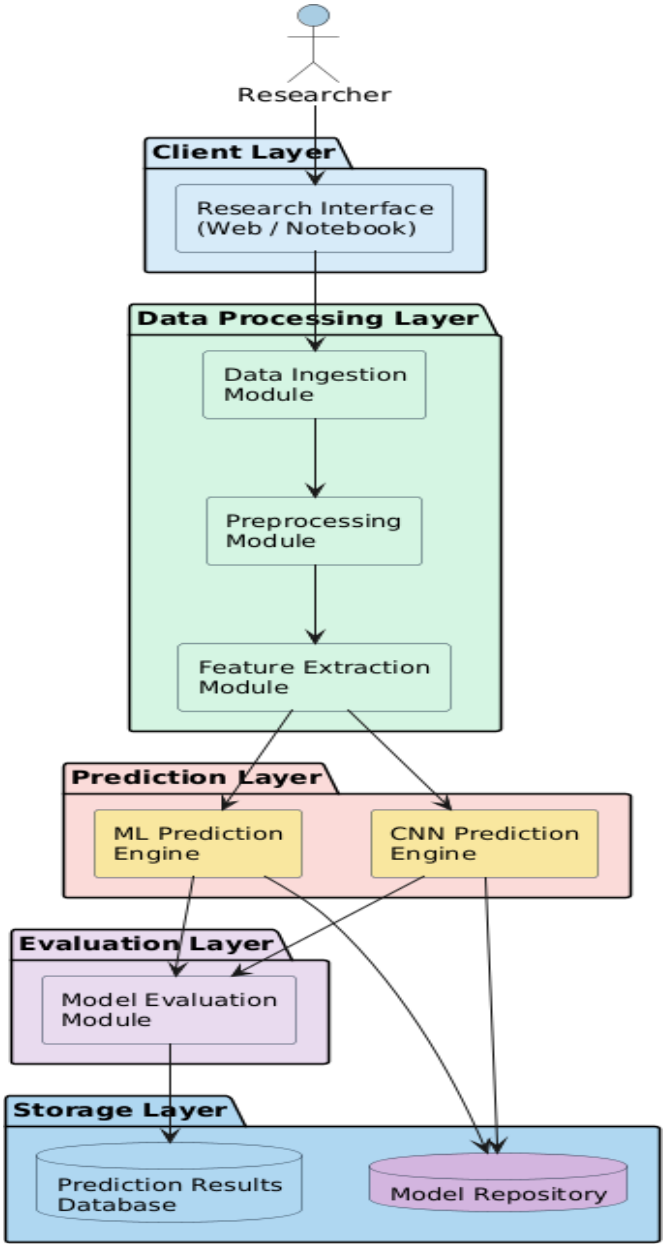


Figure 3.6 : Component Diagram for Anticancer Prediction System

3.2.3.2 System Architecture

3.2.3.3. Data Flow and Dataset Specification

Data Flow in the System:

The data flow system in Sprint II operates according to two main factors. The system uses pre-processed molecular datasets which were created during previous development stages to conduct engine research work as shown in Figure 3.6. The system uses multiple feature engineering techniques to organize data into its final format which will become the foundation of model development work.

The system creates feature-engineered datasets through various feature selection methods which include Mutual Information (MI-400) and Alternative Logistic-Random Forest (ALT-400) and Principal Component Analysis (PCA-400) as feature selection methods. The developed datasets serve as input data which machine learning and deep learning models use to conduct their training and evaluation process.

The system begins its hyperparameter optimization process after finishing model training by using the Optuna framework as its main tool. The system uses performance indicators to improve the model configurations through an iterative process. The optimized models get evaluated and saved for future applications which will happen during both ensemble learning and final prediction phases.

Datasets Used:

- 2D Descriptor Dataset:
  The dataset includes molecular descriptors which describe the structural and physicochemical characteristics of different compounds.
- Hybrid Dataset:
  The dataset combines 2D descriptors with molecular fingerprints to deliver both structural data and pattern recognition capabilities.

Derived Feature Sets:

- MI-400 Dataset:
  The dataset uses Mutual Information to select features which show their statistical value in relation to the target labels.
- ALT-400 Dataset:
  The dataset uses L1-regularized Logistic Regression together with Random Forest importance ranking to select its features.
- PCA-400 Dataset:
  The dataset applied Principal Component Analysis to its features for the purpose of achieving dimensionality reduction while maintaining variance.

Model Outputs:

- Trained Machine Learning Models:

The top-performing models were chosen based on their ability to detect all instances and their overall accuracy.

- Deep Learning Models:
  The Deep MLP and FT-Transformer and TabNet models achieved better performance through training on different datasets and feature sets.
- Optimized Models:
  The models underwent refinement through Optuna-based hyperparameter tuning which resulted in better performance and increased generalization capabilities.

Mode of Processing:

- The system executes its functions through Python-based machine learning and deep learning frameworks which include Scikit-learn and PyTorch.
- The system processes data while training models and optimizing them through a sequential pipeline.
- Optuna automates the process of hyperparameter optimization by using its system to traverse different parameter configurations.

3.2.4 Outcome of Objective

The research team for Sprint II used various methods to predict cancer drug effectiveness by developing machine learning algorithms which they tested through hyperparameter tuning. The team started this phase by choosing effective machine learning models which they tested through their advanced deep learning models which they turned into better system performance through their optimization work. The system now holds optimized models which can process advanced molecular information to enhance prediction accuracy during the next phase of ensemble learning.

3.2.4.1 Selection of High-Performing Machine Learning Models

The process to evaluate and select machine learning models based on their ability to recall information and their accuracy results succeeded. The top-performing models were identified and retained for further analysis, ensuring that only reliable models were considered. The system achieved major efficiency gains because this process eliminated duplicate elements while concentrating on the best quality models.

3.2.4.2 Training of Deep Learning Models

The research team successfully built and trained deep learning models which included Improved Deep Multilayer Perceptron (MLP) and FT-Transformer and TabNet across their various datasets and feature tracks. The models successfully learned to identify complex nonlinear patterns which existed in molecular descriptor data, resulting in better prediction results.

3.2.4.3 Hyperparameter Optimization

The Optuna framework conducted hyperparameter optimization to improve model performance. The system automatically adjusted parameters which included learning rate and batch size and number of layers and regularization techniques. The deep learning models achieved better generalization abilities while their overfitting problems were successfully reduced.

3.2.4.4 Multi-Dataset and Feature Track Evaluation

The researchers conducted training and testing of all models using multiple datasets and different feature tracks to demonstrate that their models could maintain consistent performance across all testing conditions. The method allowed for equitable model evaluation while demonstrating how different data formats affected model performance as shown in Figure 3.7.

3.2.5 Sprint Retrospective

	Sprint Retrospective			
What went well	**What went poorly**	**What ideas do you have**	**How should we take action**	
*This section highlights the **successes and positive outcomes from the sprint**. It helps the team recognize achievements and identify practices that should be*	*This section identifies the **challenges, roadblocks, or failures encountered during the sprint**. It helps pinpoint areas that need improvement or change.*	*This section is for brainstorming **new approaches, tools, or strategies to enhance the team's efficiency, productivity, or project outcomes.***	*This section outlines specific steps or solutions to address the issues and implement the ideas discussed, ensuring continuous improvement in future sprints.*	*Guidelines*
The selection of top machine learning models based on recall and accuracy improved the reliability of the system.	Some models performed inconsistently across different feature tracks and datasets.	Improve model stability by selecting models based on multiple metrics and cross-validation.	Use stratified validation and consider additional evaluation metrics for better model selection.	Filtering models based on recall ensured better identification of active anticancer compounds.
Deep learning models (Improved MLP, FT-Transformer, TabNet) were successfully trained and captured complex relationships in data.	Training deep learning models required high computational resources and longer training time.	Optimize training efficiency using better hardware utilization and early stopping techniques.	Apply early stopping, batch optimization, and efficient training strategies.	Deep learning models showed improved performance over traditional machine learning models.
Hyperparameter optimization using Optuna significantly improved model performance and generalization.	Hyperparameter tuning increased experimentation complexity and required careful tracking of results.	Automate experiment tracking and use systematic tuning approaches.	Use structured logging and experiment tracking tools to manage multiple runs.	Optuna optimization improved accuracy and reduced overfitting in deep learning models.
Training across multiple datasets and feature tracks improved robustness and model comparison.	Managing multiple datasets and feature pipelines increased system complexity.	Standardize data pipelines and maintain consistent data processing workflows.	Create unified data pipelines for all feature tracks and datasets.	Multi-dataset training enabled better generalization across different feature representations.
Integration of the healthcare platform demonstrated practical usability of the system.	Integration between AI models and frontend/backend required additional effort and coordination.	Improve integration by designing modular APIs for model access.	Develop clear interfaces between model outputs and application layer.	The platform demonstrated how predictions can be accessed in a real-world environment.

Figure 3.7: Sprint Retrospective of Anticancer Prediction System

3.3 SPRINT III – Visualization and Evaluation

3.3.1 Objective with User Stories of Sprint III

In Sprint III we will attempt to boost the prediction accuracy, stability and practical usefulness of the proposed system by using state-of-the-art model tuning techniques, massively large-scale ensembles, and the final system integration. During this sprint we will continue to work on the machine learning and deep learning pipelines we built in the previous sprints, and we will concentrate on improving model generalisation, prediction accuracy and integrating models to make a final decision.

In this sprint, we will focus on tuning hyperparameters of machine learning models using Random Search and Grid Search. These methods help us to search the hyperparameter space quicker and get the best combination to improve the performance of our models on different data sets. We will ensure that the models we select are tuned and robust enough to be well generalized on independent data.

All the deep learning models are also validated with a testing dataset, to ensure the stability of the predictions. This ensures that the models' performance can be properly assessed on unseen data and that the most consistent models are identified.

In order to enhance the prediction performance, various ensemble strategies are used. The ensemble strategies include the ensemble of deep learning models, the ensemble of machine learning models, and the ensemble of hybrid models which combine the machine learning and deep learning models. The ensemble strategies exploit the diversity of the models for more accurate and robust predictions. The system explores the different ensemble strategies with more than seventy models, which allows us to explore the model diversity and prediction performance.

Further, a convolutional neural network is developed to detect diabetic retinopathy from retinal fundus images, which enhances the health care component of the system. This model adds the image-based diagnostics to the anti-cancer prediction system.

This sprint of the sprint development cycle ends with the selection of top models based on performance metrics as shown in Table 3.4 and Figure 3.8 and 3.9. The final models are a combination of deep learning and machine learning models, which are further validated using an external set of real drug compounds. This step helps ensure practical applicability, and minimises false negatives, thus saving time and money in the subsequent real drug discovery experiments.

Finally, all parts of the system (predictive models, ensemble predictions and healthcare features) are brought together in a single, integrated web platform. This allows real-time inference, visualisation and interaction, and transforms the system from a research project into a working system.

Table 3.4: Detailed User Stories of sprint 3 for Neuro optic

S.NO	Detailed User Stories
#US 19	Hyperparameter optimization for ML both random search and grid search
#US 20	Validating all the DL models using testing dataset
#US 21	Performing 8 ensemble techniques for 12 DL models using the saved artifacts of tuning
#US 22	Performing 8 ensemble techniques for top 10 ML models using the saved artifacts of tuning

#US 23	Performing 8 ensemble techniques between ML and DL models which encomposed of another 70+ model
#US 24	Training a CNN model for predicting diabetics in human eye
#US 25	Selecting the top model which encomposed of 2 DL and 1 ML which produced excellent result and testing with external validation dataset which encomposed of authentic drug compounds saving lot of time and funds for wasting money on non active drug compounds
#US 19	Hyperparameter optimization for ML both random search and grid search

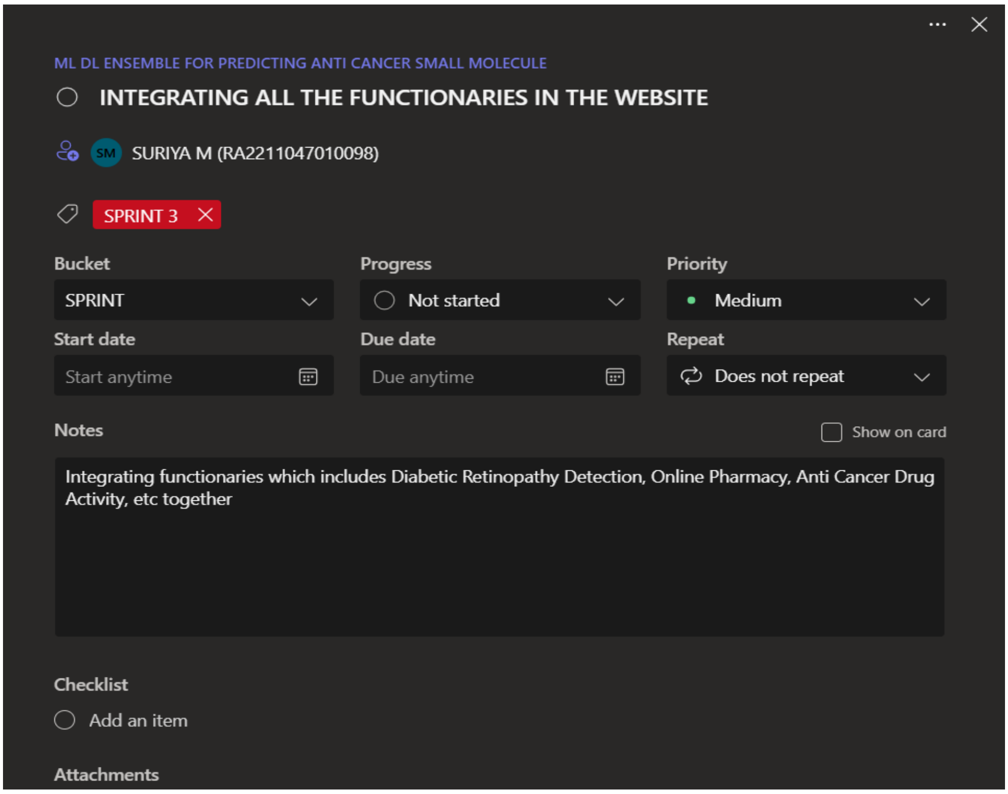


Figure 3.8 User story for Integrating all the functionaries

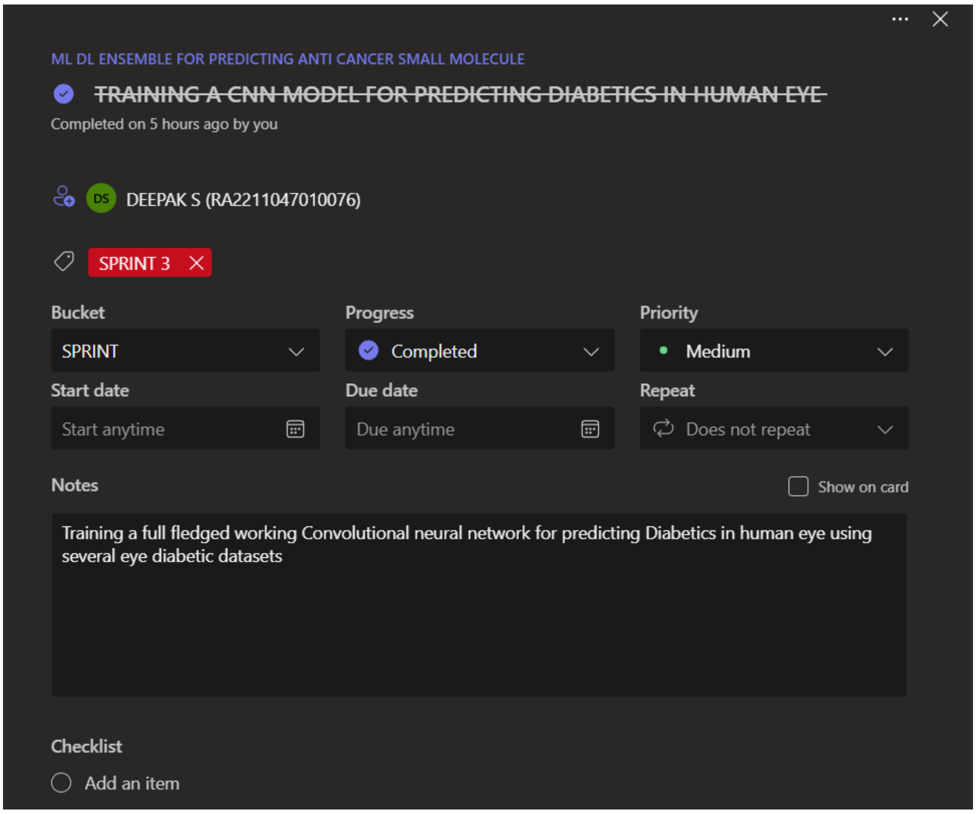


Figure 3.9 User story for Implementing CNN for Diabetic Detection

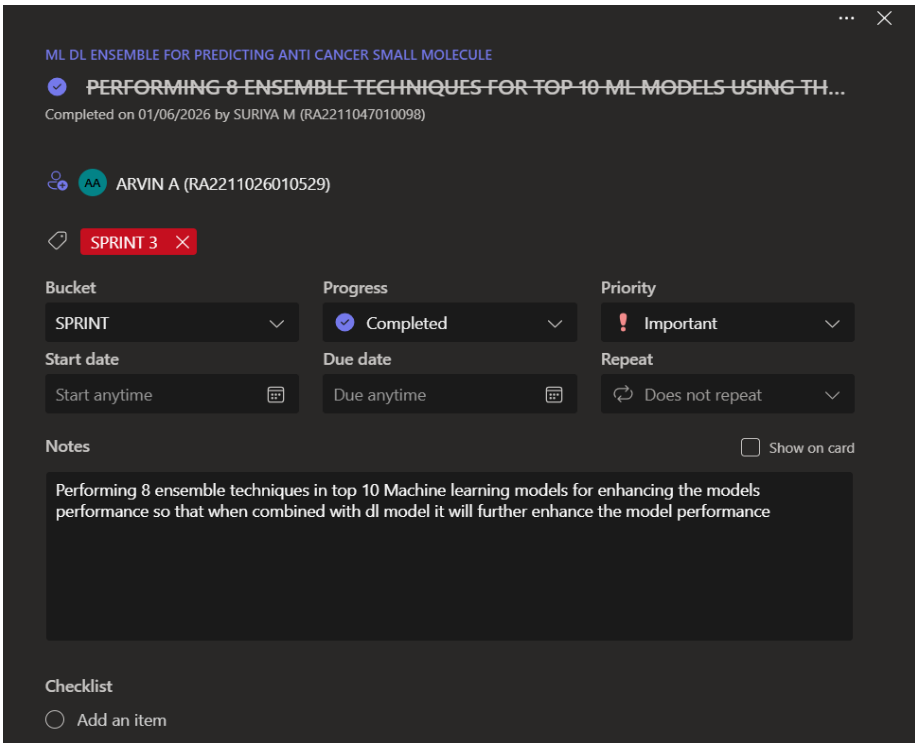


Figure 3.10 User story for Performing ensemble techniques in top 10 ML models for predicting anti-cancer activity

3.3.2 Functional Document

3.3.2.1. Introduction

This sprint is focused on the improvement of the predictive performance and robustness of the proposed system with advanced optimisation, ensemble learning and system integration. This sprint is dedicated to the optimization of hyperparameters of the developed system, large-scale model ensemble methods and internal and external validations, using the machine learning and deep learning pipelines of the previous sprints as shown in Use case Figure 3.11.

The sprint aims to increase the performance and stability of the anti-cancer drug prediction, and to enhance the health care module with the diabetic retinopathy detection by convolutional neural networks. Furthermore, the integration of the different modules in a web platform guarantees the system can be used in real-time and in real-life applications.

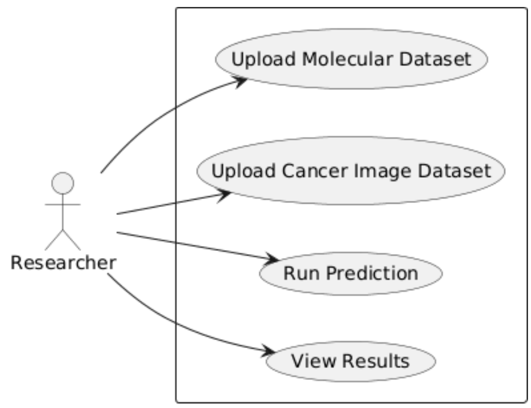


Figure 3.11: Use Case Diagram

3.3.2.2. Product Goal

In this sprint, we will build a high performance, robust and scalable AI system by tuning the model configurations, ensemble prediction using novel ensemble learning approaches and validate the models on internal and external data sets. This sprint aims to transform a set of models into a predictive system to address complex biomedical problems.

The sprint aims to:

• Fine-tune model parameters for optimal performance.

• Test model accuracy and generalizability with test data.

• Improve prediction performance and stability through ensemble learning.

• Integrate machine learning and deep learning models into a decision-making system.

• Experiment with models to determine optimal settings.

• Validate models on other data sets to assess their generalizability.

• Extend the health module by CNN image analysis.

• All features should be available in a user-friendly web platform.

3.3.2.3. Demography

Users:

• Biomedical researchers in drug design and molecular research.

• AI and data scientists working on prediction and optimization.

• Medical practitioners seeking diagnostic assistance.

• Drug researchers screening for anti-cancer drugs.

• General public using AI-based diagnostic and health-care software.

Location:

• Universities and research institutes in AI and biomedical research.

• Medical facilities requiring early diagnosis.

• Drug discovery and development environments.

• Internet-based and web-accessible systems.

• Multiregional and multi dataset scalability.

3.2.2.4. Business Processes

Model Optimization

- Random Search and Grid Search are used to tune the ML models' hyperparameters
- The best combinations are selected using metrics

Model Validation

- Deep learning models are tested on a separate testing set
- It is ensured that the performance is consistent

Ensemble Modelling

- We use different types of ensemble for:
    - Deep learning models
    - Machine learning models
    - Ensembles with combinations of ML and DL models

- The ensemble responses are predicted using model artifacts

Healthcare Model Development

- A CNN is developed to diagnose diabetic retinopathy
- The model accepts fundus images of the retina and produces diagnostic labels

External Validation

- The best models are chosen from the performance metrics
- These models are tested on external test sets of real drug molecules
- This guarantees applicability and reduces false positives

System Integration

- The system is web-based with all modules integrated
- Backend services are used for model inference and data processing
- Frontend interfaces provide visualization and user interaction

3.3.2.5 Features

Feature 1: Hyper-parameter tuning

Description: This feature enhances the accuracy of the machine learning models by optimising the hyper-parameters of models using random search and grid search techniques to determine the optimal hyper-parameters combinations.

User Story:

I need to tune the parameters of the models in order to increase the accuracy of the predictions.

Feature 2: Test deep learning models

Description: Deep learning models are validated on a test dataset to ensure the model is stable, robust and generalises well to unseen data.

User Story:

I want to be a data scientist and validate models on unseen data so that I can be confident.

Feature 3: Ensemble of Deep Learning Models

Description:

Several deep learning models are trained and combined through ensemble methods to enhance model performance and robustness.

User Story:

I want to ensemble deep learning models so that I can boost my predictions.

Feature 4: Ensemble of Machine Learning Models

Description:

Machine learning models with the best performance are used in an ensemble framework to improve the overall prediction accuracy and stability.

User Story:

I want to ensemble machine learning models so that I can enhance model performance.

Feature 5: Ensemble of ML and DL Models

Description:

This feature allows the ensemble of machine learning and deep learning models, so that the ensemble can take the advantage of different algorithms.

User Story: As a developer, I want to ensemble ML and DL models so that I can get better generalization.

Feature 6: Scalable Ensemble Architecture

Description:

It supports large-scale ensemble combinations, with multiple models and configurations in the ensemble, to support large-scale experimentation and tuning.

User Story: As a researcher I want to try various ensemble combinations, so that I can select the best model.

Feature 7: CNN-based Diabetic Retinopathy Detection

Description: It is trained to use convolutional neural network to analyze retinal fundus images and detect diabetic retinopathy, which provides image-based diagnostic ability.

User Story: As a user I want to upload my retinal images for early diagnosis

Feature 8: Model ranking and selection

Description:

Models are ranked using various metrics such as accuracy, recall and generalization to identify the top models.

User Story: As a researcher, I want to be able to rank models so that I can choose the best ones.

Feature 9: Validation on External Datasets

Description: The best models are then validated using external data sets that comprise real drug compounds to ensure that they can be used.

User Story:

I want to be able to test models on external data as a researcher so that I can ensure that they work in practice.

Feature 10: Reuse Models from Artifacts

Description:

Model artifacts from prior training and tuning stages are reused for model ensemble, speeding up the process and avoiding inconsistencies.

User Story: As a developer, I want to reuse models so that I don't have to retrain them.

Feature 11: Multimodal System Integration

Description: Link up the molecular prediction models, medical image models and user interface modules.

User Story:

I as a user, want to access all the functionality in one system for convenient use.

Feature 12: Prediction and Inference in Real-Time

Description:

Real-time predictions are available on the platform for both anti-cancer molecules and diagnostics.

User Story:

As a user, I want predictions in a timely manner.

Feature 13: Interactive Web-Based Interface

Description:

An interactive web-based environment allows users to interact with models, input data and view results.

User Story:

I want a simple interface so that I don't need computer skills.

3.3.3 Architecture Document

3.3.3.1 Application

The Sprint III architecture is designed to be a scalable and modular system that integrates health services, anti-cancer prediction and user interaction features. The architecture adopts a microservices approach, in which each component is responsible for a single function, allowing the system to be scalable, flexible and transferable is clearly demonstrated in Figure 3.12.

The key architectural components are outlined below:
User Interaction Layer:

This layer provides the interface for the user to interact with the system, through the web portal. The input can be text, speech and visual data.

- Speech Input Module: This module can record user's voice via a microphone and converts it to text via speech recognition.
- Web Interface (Frontend): This module is built using contemporary web technologies to offer an interactive platform for user interaction, data visualization and data input.

Conversational AI Module:
This is responsible for user queries and mental health coaching.

- Asks questions using Natural Language Processing Generates responses using language models Provides interactive and contextual conversation Facilitates text and speech interaction
- Speech Processing Module Speech-to-Text: Converts user's speech into text
- Text-to-Speech: Translates responses into speech Enables real-time conversation

Backend Processing Layer:

- The backend is a Flask-based architecture which is the processing engine of the system.

- Machine Learning and Deep Learning Module Receives API calls from the frontend. Processes user input and passes it to other modules Interfaces with AI models and other swrvices.
- Manages the inference pipelines for machine learning and deep learning models.

Machine Learning and Deep Learning Module:
This is the core processing engine of Sprint III.

- Has tuned machine learning models (XGBoost, LightGBM, etc.)
- Includes deep learning models (MLP, Transformer, CNN)
- Hyperparameter tuning (Random Search and Grid Search)
- Large-scale ensemble techniques:
  - ML ensembles
  - DL ensembles
  - Hybrid ML-DL ensembles
- Fast predictions by using model artifacts
- Predictions for anti-cancer molecules

CNN-Based Healthcare Module:

- Retinal fundus image processing
- Deep learning for the diagnosis of diabetic retinopathy Diagnostic output incorporated into system

External Validation:

- Module Validation of the top models with external data sets
- Real-world prediction validation Adds confidence in drug discovery process
- Database Integration Layer Storage of user data Storage of model outputs Interaction logs Structured data (e.g., SQLite)
- Efficient data retrieval and storage

API Integration Layer:

- Use of external services through REST APIs
- Google Maps (doctor finder)

- Online pharmacy services
- Real-world connectivity

Visualization and Output Layer:

- Display of predictions Visualisation of insights for users
- Results of model in a comprehensible way Easy-to-understand results.
- Animated Psychiatrist for interactive visualisation Animation with speech output Enhancement of user engagement.

3.3.3.2 System Architecture

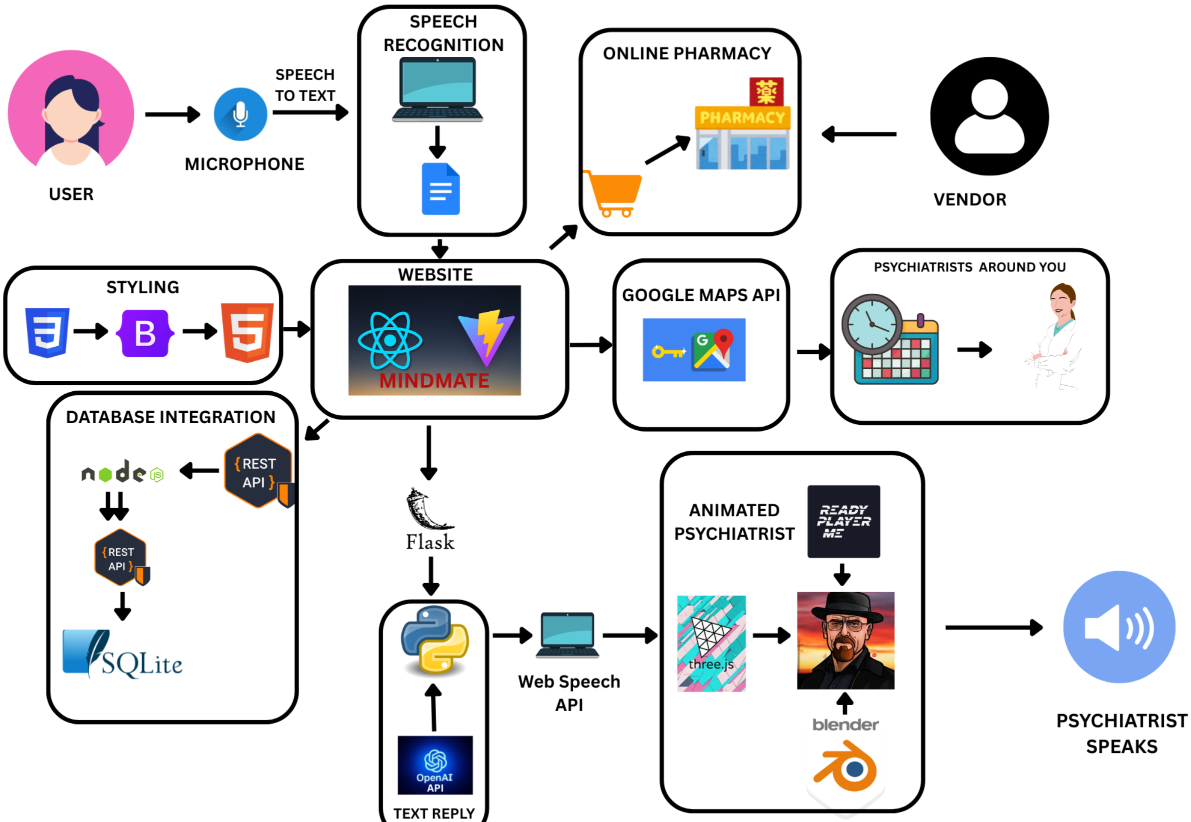


Figure 3.12 Architecture Diagram for Integration of website

The system architecture follows a multi-layered design consisting of:

1. Input Layer:
   - User Inputs (Text, speech, images)
2. Processing Layer:
   - Speech recognition

    - NLP processing
    - ML/DL inference
3. Intelligence Layer
    - Anti-cancer prediction models
    - CNN-based diagnostic models
    - Ensemble decision systems
4. Integration Layer
    - APIs (Google Maps, Pharmacy)
    - Database management
5. Output Layer
    - Predictions
    - Visualizations
    - Audio responses

Architecture enables efficient communication between modules through APIs, data pipelines, and real-time processing and scalability.

3.3.3.3 Data Exchange

Frequency of Data Exchange:

- User input is processed in real-time
- Model inference is triggered on demand
- External API calls are dynamic based on user interaction
- Database updates happen after each interaction or prediction

Data Types:

Table 3.4: Data Types

Data Type	Description
User Input Data	Text, speech, and image inputs
Molecular Data	Descriptor-based features for prediction
Model Outputs	Predictions, probabilities, ensemble results
Healthcare Data	Retinal images and diagnostic outputs
API Data	Location and service-based data

Mode of Exchange:

- REST APIs for frontend-backend communication
- JSON format for data transfer
- Python-based pipelines for model inference
- Database queries for storage and retrieval

3.3.4 Outcome of the objective

The goal of Sprint III was to improve the performance, stability, and practical usefulness of the proposed system by tuning the models, ensembling models at scale, and integrating the system. This sprint's results show substantial improvements in predictive performance, robustness, and readiness for deployment.

3.3.4.1 Machine Learning Model Hyperparameter Optimization

Objective: To enhance the accuracy of machine learning models through hyperparameter optimization using approaches like Random Search and Grid Search.

Outcome: We were able to fine-tune machine learning models by trying out different combinations of parameters, leading to enhanced model accuracy and generalization on datasets. The optimization process led to the selection of the best model configurations, improving recall and model robustness.

3.3.4.2 Testing of Deep Learning Models

Objective: To assess the deep learning models on a separate validation set.

Outcome: Deep learning models were tested on separate data to confirm their stability and performance. It was confirmed that the models could learn complex relationships between the input and output, and were stable to different variations of the dataset.

3.3.4.3 Ensemble Modeling for Deep Learning Models

Objective: To combine best performing deep learning models using ensemble approaches to enhance prediction performance.

Outcome: Ensembles of deep learning models improved prediction accuracy through variance reduction and the utilization of different deep learning model combinations. This resulted in more consistent predictions than individual models.

3.3.4.4 Ensemble Modelling for Machine Learning Models

Objective: To improve performance by combining best performing machine learning models.

Outcome: Ensemble machine learning models were identified to be more accurate by combining the predictions of several models. The ensemble approach reduced the bias of individual models and enhanced the system's performance.

3.3.4.5 Ensemble of ML and DL Models

Objective: To combine the machine learning and deep learning models into a hybrid ensemble.

Outcome: The hybrid ensemble approach, by leveraging the strengths of both types of models, resulted in enhanced prediction accuracy. This integration enabled the system to more efficiently handle different data patterns, leading to better generalization and robustness.

3.3.4.6 CNN Based Detection of Diabetic Retinopathy

Objective: To create a convolutional neural network (CNN) that can detect diabetic retinopathy.

Outcome: The CNN model was able to learn the visual patterns associated with diabetic retinopathy and was able to predict this condition from retinal fundus images. This enhanced the health-care component, offering image-based diagnosis.

3.3.4.7 External Dataset Validation

Objective: To validate the best models in an external dataset of real drug compounds.

Outcome: The top performing models were validated using external data and demonstrated good performance. This confirmed the feasibility of the anti-cancer prediction system and reduced the chances of prediction errors, thus aiding the process of drug discovery.

3.3.4.8 System Integration and Deployment

Objective:
To integrate all system components into a unified web-based platform.

Outcome: The system was integrated all the system components (predictive models, ensemble systems, healthcare system and user interaction modules) in one system. The system is real-time inference, visualisation and interactive system thus turning framework into a user-friendly application.

3.3.5 Sprint Retrospective

	Sprint Retrospective			
What went well	**What went poorly**	**What ideas do you have**	**How should we take action**	**Examples**
*This section highlights the **successes and positive outcomes from the sprint**. It helps the team recognize achievements and identify practices that should be continued.*	*This section identifies the **challenges, roadblocks, or failures encountered during the sprint**. It helps pinpoint areas that need improvement or change.*	*This section is for brainstorming **new approaches, tools, or strategies to enhance the team's efficiency, productivity, or project outcomes.***	*This section outlines specific steps or solutions to address the issues and implement the ideas discussed, ensuring continuous improvement in future sprints.*	*Guidelines*
Hyperparameter optimization using Random Search and Grid Search significantly improved model performance and stability.	Tuning process was computationally expensive and time-consuming due to large parameter space.	Introduce more efficient optimization strategies such as Bayesian optimization or early stopping techniques.	Optimize search space, apply resource-efficient tuning, and parallelize computations where possible.	Optimized models showed improved recall and generalization across datasets.
Validation of deep learning models ensured reliability and consistent performance on unseen data.	Some models showed slight overfitting due to dataset limitations.	Incorporate stronger regularization and data augmentation techniques.	Improve dataset diversity and apply dropout, augmentation, and early stopping.	Validated models maintained stable performance across testing datasets.
Ensemble techniques across DL models improved prediction robustness and reduced variance.	Managing multiple DL models increased complexity in implementation and inference time.	Explore weighted ensembles and model pruning techniques.	Reduce redundant models and optimize ensemble selection strategies.	DL ensembles provided more stable predictions than individual models.
Ensemble techniques for ML models enhanced accuracy and reduced individual model bias.	Some models contributed minimally to ensemble performance.	Apply model selection techniques to retain only impactful models.	Use feature importance and performance metrics to filter models.	ML ensembles improved overall prediction consistency.
Hybrid ML-DL ensemble integration produced superior results by combining complementary strengths.	Integration required normalization of outputs and handling different prediction formats.	Standardize output pipelines for all models.	Implement unified prediction formats and scaling mechanisms.	Hybrid ensemble achieved better generalization than standalone approaches.
CNN model for diabetic retinopathy detection successfully captured image-based patterns.	Model performance depended on image quality and dataset size.	Improve dataset quality and use transfer learning techniques.	Incorporate pre-trained models and enhance preprocessing.	CNN enabled accurate detection of retinal abnormalities.
External dataset validation confirmed real-world applicability of selected models.	Slight performance variation observed between internal and external datasets.	Increase diversity of training data and perform domain adaptation.	Expand dataset coverage and improve model robustness.	External validation reduced risk of false predictions in drug discovery.
Integration of all modules into a web platform enabled real-time interaction and usability.	Integration complexity increased due to multiple modules and dependencies.	Modularize system further and improve API efficiency.	Refactor backend services and optimize API communication.	Users can access predictions, chatbot, and diagnostics in a single platform.

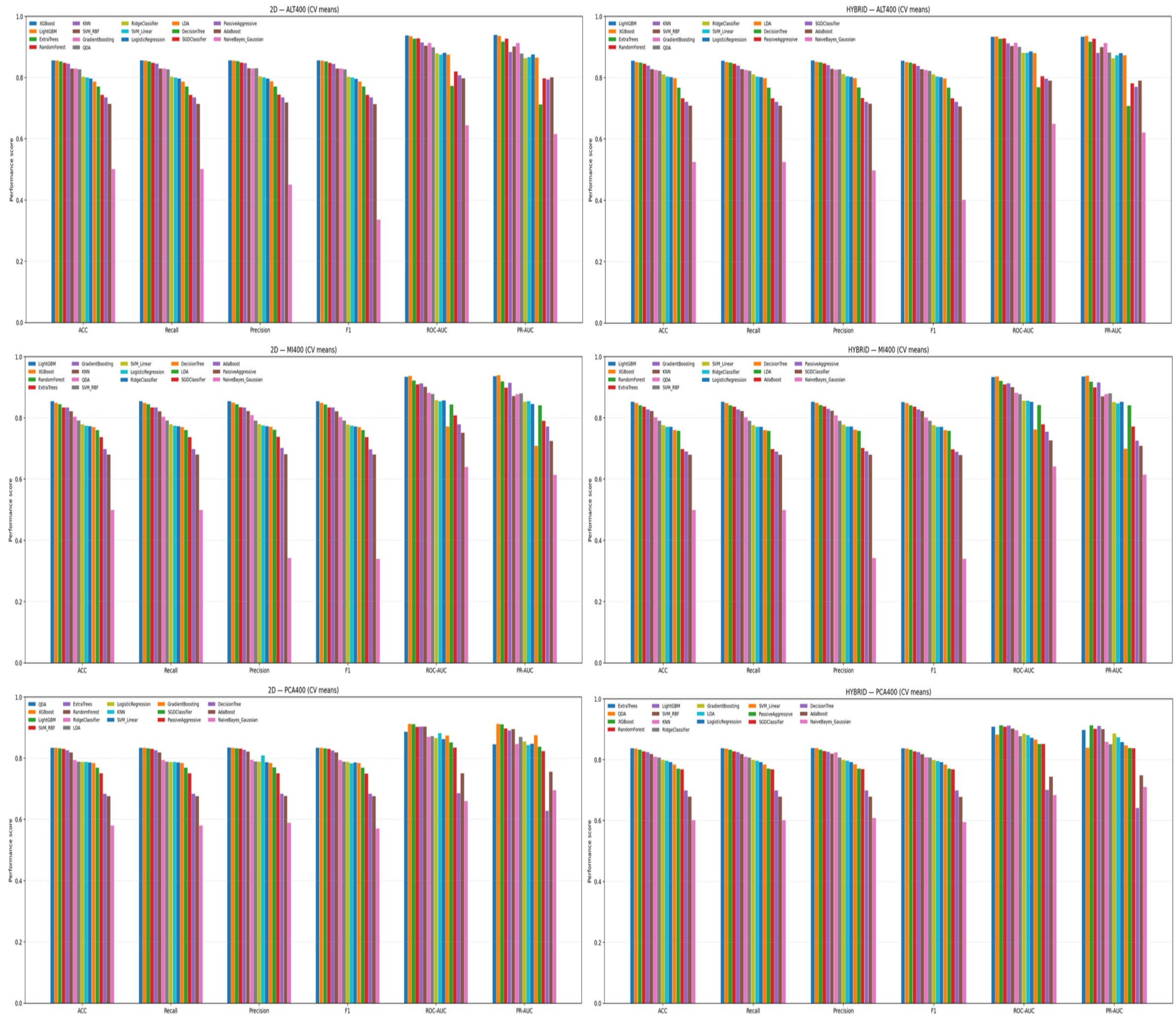
Figure 3.13: Sprint Retrospective of Neuro-Optic

CHAPTER 4

RESULTS AND DISCUSSION

4.1. Performance of Machine Learning and Deep Learning Models

4.1.1. Machine Learning Model Performance



(a) (b)

Figure 4.1. Performance comparison of 102 baseline machine learning models on 2D and Hybrid datasets using ALT400, MI400, and PCA400 features.

A total of 102 baseline ML models were trained across the ALT400, MI400 and PCA400 feature selection tracks for both the 2D and Hybrid datasets and their overall performance was visualized in Figure 4.1. The top ten models selected primarily based on Recall and ensuring diversity among ensemble families from training where 2D MI400 - XGBoost and Hybrid ALT400 - LightGBM tops the chart with 0.86 recall followed by 2D MI400 - LightGBM with 0.85 recall and other models. The top 10 models were hence evaluated with test dataset and results were given in Table 4.1 and overall performance was visualized in Figure 4.2, the Hybrid ALT400 – XGBoost model achieved the strongest overall test performance, with Accuracy = 0.77, Recall = 0.77, F1-macro = 0.76, and ROC-AUC = 0.90. Very similar performance was observed for 2D ALT400 – XGBoost (Accuracy/Recall = 0.77; ROC-AUC = 0.89) and 2D ALT400 – ExtraTrees (Accuracy/Recall = 0.77; ROC-AUC = 0.87). Overall, the top ML models demonstrated Accuracy and Recall values in the range of 0.73–0.77, Precision values of 0.77–0.82, and ROC-AUC values between 0.86–0.90, Indicating good separability between active and inactive molecules across both datasets. Notably, ensemble tree-based learners (XGBoost, LightGBM and ExtraTrees) consistently outperformed linear and kernel-based approaches, confirming their ability to capture non-linear structure in high-dimensional molecular descriptor data and providing a strong baseline prior to deep learning and ensemble integration.

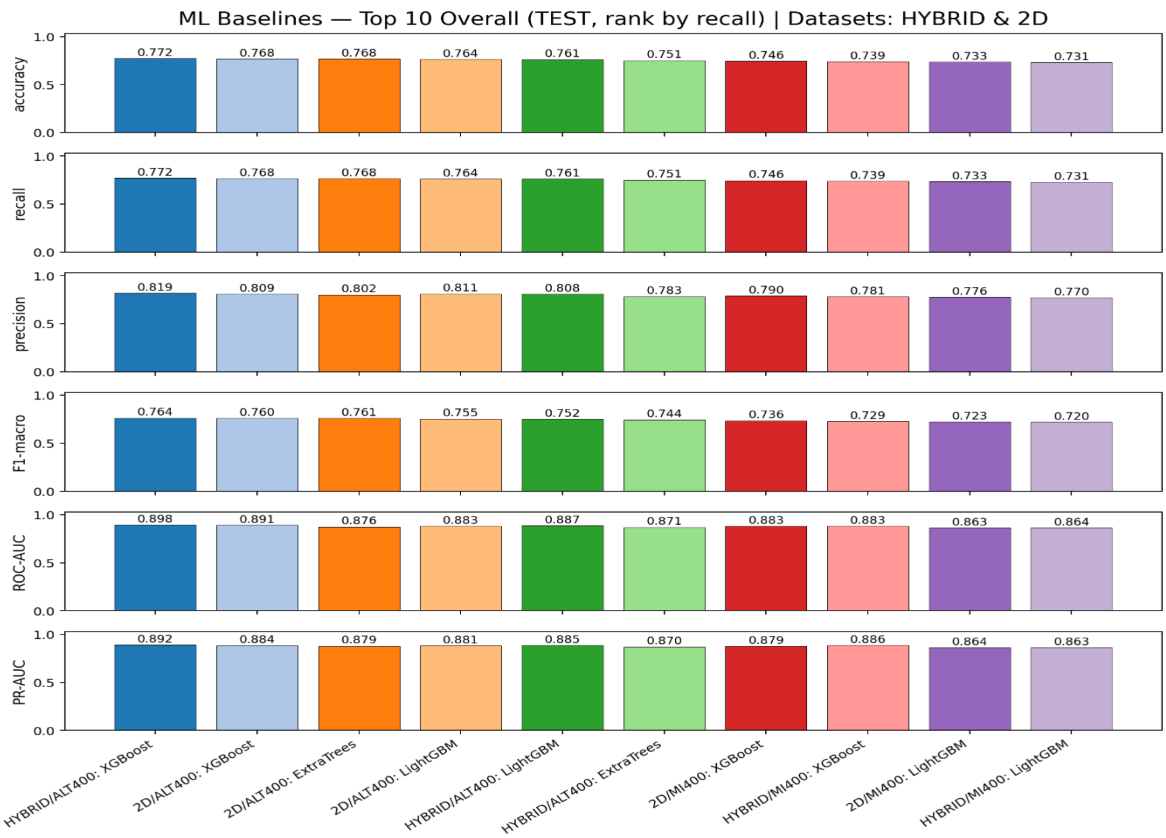


Figure 4.2. Performance comparison of top machine learning models on 2D and Hybrid datasets using ALT400 and MI400 features.

Table 4.1. Comparative performance of machine learning models on 2D and Hybrid datasets using ALT400 and MI400 features.

Dataset	Feature selection	Model	Accuracy	Recall	Precision	F1-macro	ROC-AUC	PR-AUC
2D	ALT400	XGBoost	0.77	0.77	0.81	0.76	0.89	0.88
		ExtraTrees	0.77	0.77	0.80	0.76	0.87	0.89
		LightGBM	0.76	0.76	0.81	0.75	0.88	0.88
2D	MI400	XGBoost	0.75	0.75	0.79	0.74	0.88	0.88
		LightGBM	0.73	0.73	0.78	0.72	0.86	0.86
Hybrid	ALT400	XGBoost	0.77	0.77	0.81	0.76	0.90	0.89
		ExtraTrees	0.75	0.75	0.78	0.74	0.87	0.87
		LightGBM	0.76	0.76	0.81	0.75	0.87	0.88
Hybrid	MI400	XGBoost	0.74	0.74	0.78	0.73	0.88	0.89
		LightGBM	0.73	0.73	0.77	0.72	0.86	0.86

4.1.2. Deep Learning Model Performance

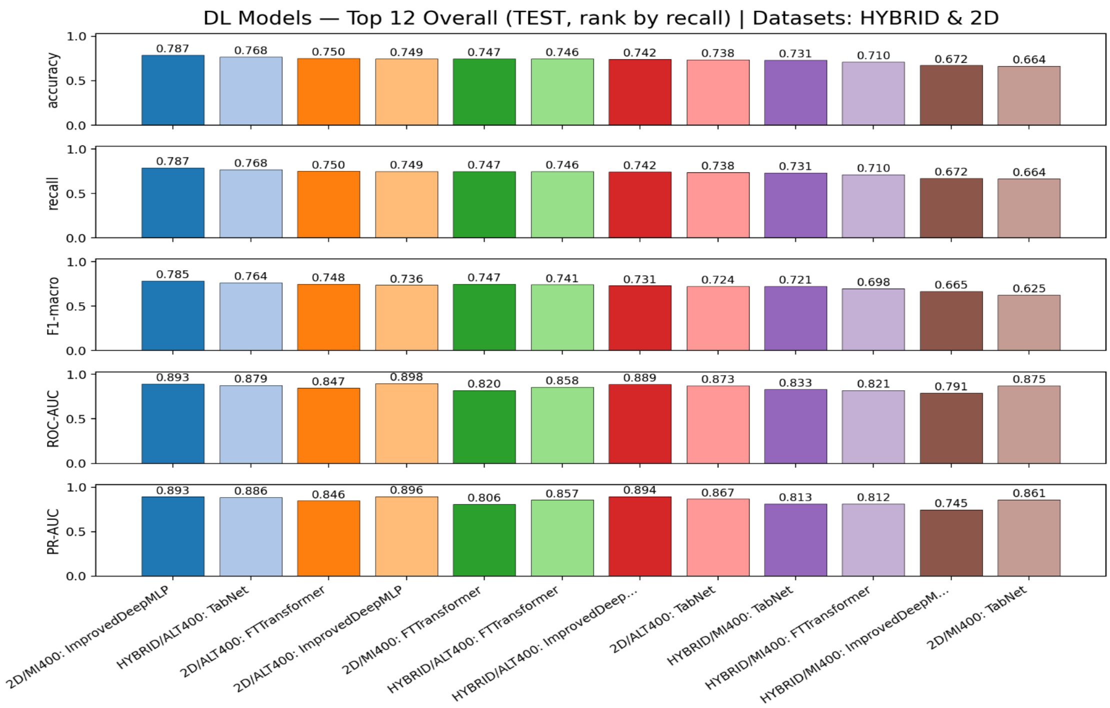


Figure 4.3. Performance comparison of deep learning models on 2D and Hybrid datasets using ALT400 and MI400 features.

Deep learning models were implemented using three architectures, Improved Deep MLP, FT-Transformer and TabNet and trained across two tracks MI400 and ALT400. Table 4.2 summarises the twelve DL models performance, and Figure 4.3 illustrates their multi-metric performance. The highest overall performance was achieved by the Improved Deep MLP model on 2D MI400, reaching Accuracy = 0.79, Recall = 0.79, F1-macro = 0.78, ROC-AUC = 0.89, and PR-AUC = 0.89. The Improved Deep MLP on 2D ALT400 obtained the highest ROC-AUC score (0.90) among all DL models, highlighting its strong discriminative ability. FT-Transformer configurations showed intermediate performance with ROC-AUC ranging approximately between 0.82–0.89, demonstrating effective modelling of feature dependencies. TabNet, although competitive in certain tracks (e.g., 2D ALT400 with Accuracy ≈ 0.75), also produced the lowest scores among all DL models in some configurations (down to Accuracy = 0.66 and ROC-AUC = 0.79), indicating higher variability compared to the other architectures. Overall, across the top twelve DL models, Accuracy and Recall varied between ~0.66–0.79, F1-macro between ~0.62–0.79, ROC-AUC between ~0.79–0.90, and PR-AUC between ~0.74–0.90, suggesting that the best DL models reach or exceed the performance of top ML baselines, while also offering enhanced interpretability through attention-based mechanisms.

Table 4.2. Comparative performance of deep learning models on 2D and Hybrid datasets using ALT400 and MI400 features.

Dataset	Feature selection	DL model	Accuracy	Recall	F1-macro	ROC-AUC	PR-AUC
2D	ALT400	TabNet	0.74	0.74	0.72	0.87	0.87
		FTTransformer	0.75	0.75	0.75	0.85	0.85
		ImprovedDeepMLP	0.75	0.75	0.74	0.90	0.90
2D	MI400	TabNet	0.66	0.66	0.62	0.87	0.86
		FTTransformer	0.76	0.76	0.75	0.82	0.81
		ImprovedDeepMLP	0.79	0.79	0.78	0.89	0.89
Hybrid	ALT400	TabNet	0.77	0.77	0.76	0.88	0.89
		FTTransformer	0.75	0.75	0.74	0.86	0.86
		ImprovedDeepMLP	0.74	0.74	0.73	0.89	0.89

Hybrid	MI400	TabNet	0.73	0.73	0.72	0.83	0.81
		FTTransformer	0.71	0.71	0.70	0.82	0.81
		ImprovedDeepMLP	0.67	0.67	0.66	0.79	0.74

4.1.3. Out-of-Fold (OOF) Performance for ML and DL

The OOF predictions of the top selected ML and DL models with the goal of improving generalization and reducing overfitting. An aggregation of probability outputs over folds has been used for ensembling, allowing it to exploit the complementary strengths of both model types. The results from OOF predictions of model were Accuracy, Recall, and F1-macro between 0.82 and 0.86, while ROC-AUC and PR-AUC scores varied between 0.90 and 0.94, reflecting outstanding discriminative ability between active and inactive compounds. Among base models, LightGBM and XGBoost yielded the best individual contributions (Accuracy around 0.86, ROC-AUC around 0.94) followed by TabNet and Improved Deep MLP.

4.2. Ensemble Performance

4.2.1. Machine Learning Ensembles

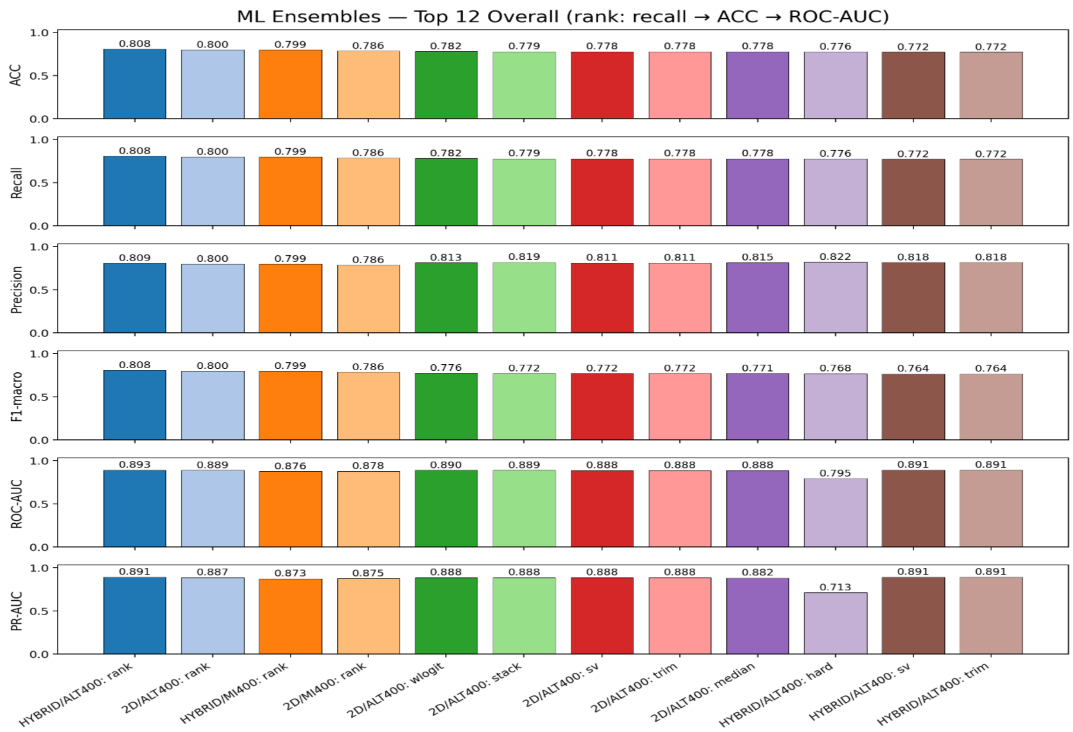


Figure 4.4. Performance comparison of ensemble machine learning models on 2D and Hybrid datasets using ALT400 and MI400 features.

A total of 10 ML models were used for ensemble across the ALT400 and MI400 feature tracks for both the 2D and Hybrid datasets. The top ensemble results are summarized in **Table 4.3** and **Figure 4.4** using multiple metrics which includes Accuracy, Recall, Precision, F1-macro, ROC-AUC, and PR-AUC. According to the ensemble results, the highest-performing model was Hybrid ALT400 – rank, which achieved Accuracy = 0.81, Recall = 0.81, F1-macro = 0.81, and ROC-AUC = 0.89, followed by 2D ALT400 – rank (Accuracy and Recall = 0.80, ROC-AUC = 0.89) and Hybrid MI400 – rank (Accuracy and Recall = 0.80, ROC-AUC = 0.88). Across the top 12 ML-ensemble configurations, Accuracy and Recall ranged between ~0.78–0.81, Precision between ~0.79–0.82, F1-macro between ~0.76–0.81, ROC-AUC between ~0.80–0.89, and PR-AUC between ~0.71–0.89, reflecting strong separability between active and inactive molecules with comparatively low false-positive rates. Overall, the ML-ensemble results clearly show that rank-based boosting ensembles such as LightGBM, XGBoost, and ExtraTrees consistently outperform linear and kernel-based models and provide a strong baseline (0.81 Accuracy and Recall) prior to applying deep learning and hybrid ensemble integration.

Table 4.3. Comparative performance of ensemble machine learning models on 2D and Hybrid datasets using ALT400 and MI400 features.

Data set	Feature selection	Ensemble	ML model	Accuracy	Recall	Precision	F1-macro	ROC-AUC	PR-AUC
2D	ALT400	rank_average_ml_all	ExtraTrees-LightGBM-XGBoost	0.80	0.80	0.80	0.80	0.89	0.89
		weighted_logit_ml_all		0.78	0.78	0.81	0.78	0.89	0.89
		stacking_strong_ml_all		0.78	0.78	0.82	0.77	0.89	0.89
		soft_vote_equal_ml_all		0.78	0.78	0.81	0.77	0.89	0.89
		trimmed_mean_ml_all		0.78	0.78	0.81	0.77	0.89	0.89
		median_vote_ml_all		0.78	0.78	0.81	0.77	0.89	0.88
2D	MI400	rank_average_ml_all	LightGBM-XGBoost	0.79	0.79	0.79	0.79	0.78	0.87
Hybrid	ALT400	rank_average_ml_all	ExtraTrees-LightGBM-	0.81	0.81	0.81	0.81	0.89	0.89
		hard_vote_ml_all		0.78	0.78	0.78	0.82	0.77	0.71

		soft_vote_equal_ml	XGBoost	0.77	0.77	0.82	0.76	0.89	0.89
		trimmed_mean_ml_all		0.77	0.77	0.82	0.76	0.89	0.89
Hybrid	MI400	rank_average_ml_all	LightGBM-XGBoost	0.80	0.80	0.80	0.80	0.88	0.87

4.2.2. Deep Learning Ensembles

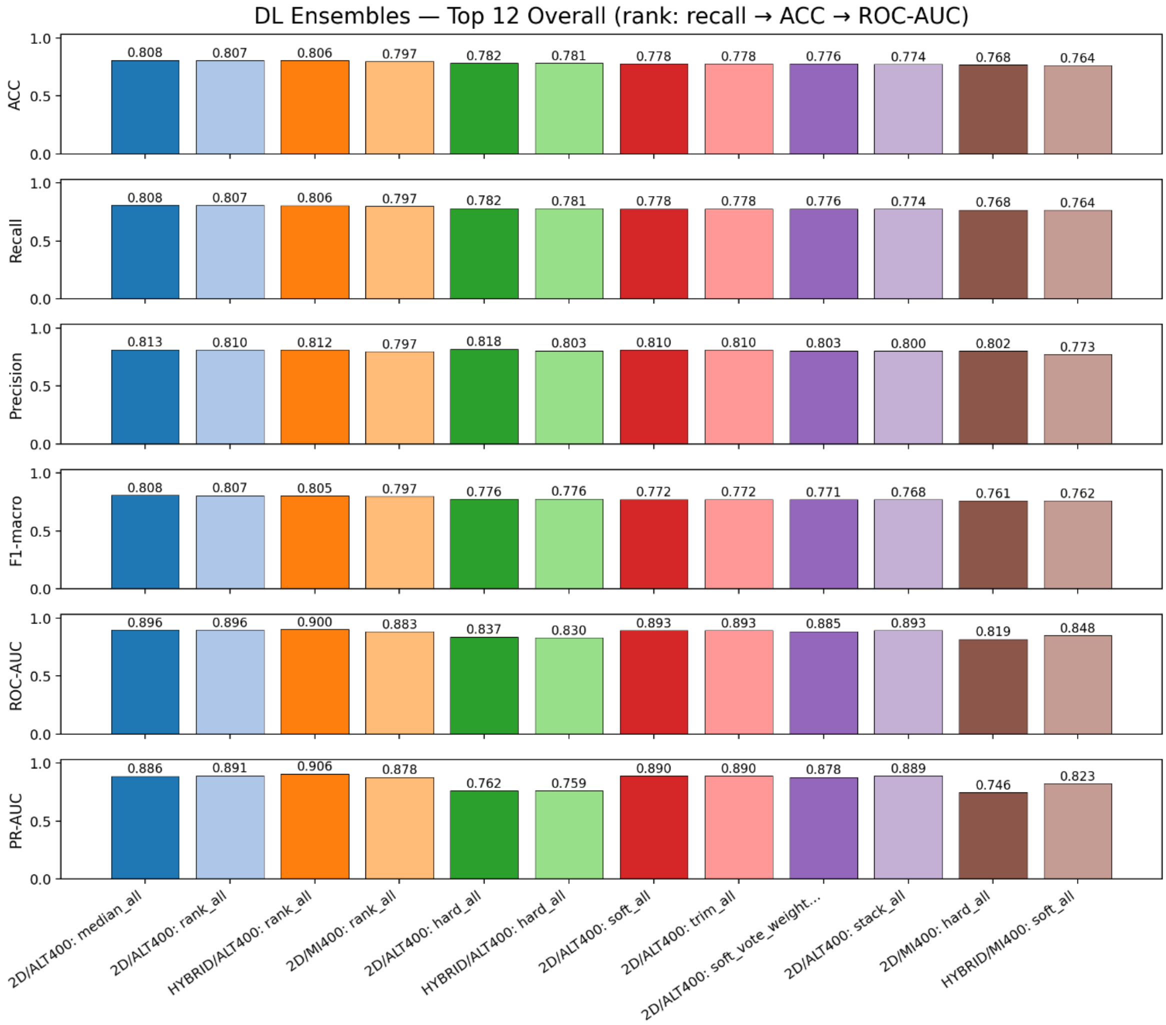


Figure 4.5. Performance comparison of ensemble deep learning models on 2D and Hybrid datasets using ALT400 and MI400 features.

Deep learning ensembles, combining Improved Deep MLP, FT-Transformer, and TabNet, were assessed on both 2D and Hybrid datasets using multiple fusion strategies, including soft voting, median aggregation, trimmed-mean fusion, and rank averaging. The top DL ensemble results are summarized in **Table 4.4** and visualized in **Figure 4.5**. Across all configurations, the ensembles achieved Accuracy and Recall in the range of 0.77–0.81, with ROC-AUC values between 0.81–0.90 and PR-AUC values between 0.76–0.91, indicating reliable and stable predictive performance across feature tracks. The 2D ALT400 – median_all ensemble which consists of all the 12 DL models achieved the strongest overall results with Accuracy = 0.81, Recall = 0.81, ROC-AUC = 0.90, and PR-AUC = 0.89, closely followed by 2D ALT400 – rank_all with Accuracy and recall values of 0.80 and ROC-AUC values of 0.90. These results confirm that the DL ensemble strategies reduce variance and improve generalization compared to individual models by leveraging complementary learning behaviours across different neural architectures.

Table 4.4. Comparative performance of ensemble deep learning models on 2D and Hybrid datasets using ALT400 and MI400 features.

Data set	Feature selection	Ensemble	DL model	Accuracy	Recall	Precision	F1-macro	ROC-AUC	PR-AUC
2D	ALT400	rank_dl_all	FTT-ImpDMLP-TabNet	0.81	0.81	0.81	0.81	0.90	0.89
		median_dl_all		0.80	0.80	0.81	0.81	0.90	0.89
		stack_dl_all		0.77	0.77	0.80	0.77	0.89	0.89
		soft_dl_all		0.78	0.78	0.81	0.77	0.89	0.89
		trim_dl_all		0.78	0.78	0.81	0.77	0.89	0.889
		soft_vote_weighted_dl_all		0.78	0.78	0.80	0.77	0.88	0.88
		hard_dl_all		0.78	0.78	0.82	0.78	0.84	0.76
2D	MI400	rank_dl_all		0.80	0.80	0.80	0.80	0.88	0.88
		hard_dl_all		0.77	0.77	0.80	0.76	0.82	0.75
Hybrid	ALT400	rank_dl_all		0.81	0.81	0.81	0.80	0.90	0.90
		hard_dl_all		0.78	0.78	0.80	0.78	0.83	0.76
Hybrid	MI400	soft_dl_all		0.76	0.76	0.77	0.76	0.85	0.82

4.2.3. Hybrid ML-DL Ensembles

Figure 4.6. Performance comparison of ensemble machine learning and deep learning models.

Hybrid ensembles, combining the best ML models (XGBoost, LightGBM, ExtraTrees) with the best DL architectures (Improved Deep MLP, FT-Transformer, TabNet), were constructed using different fusion methods such as rank averaging, median aggregation, soft and hard voting, stacking, and trimmed-mean fusion. The performances of these ensembles were evaluated on both the 2D and Hybrid datasets for both the ALT400 and MI400 feature selection methods were given in **Table 4.5.** and visualized in **Figure 4.6.** The combined ML-DL ensembles achieved Accuracy, Recall and F1-macro values ranging between 0.78 and 0.83, with ROC-AUC scores of 0.89-0.90 and PR-AUC values of 0.89-0.91 indicating strong and consistent discriminative performance. The overall best results were obtained by the 2D MI400 ensemble, Rank-DLML-Top3: (Accuracy = 0.83, Recall = 0.83, F1-macro = 0.83, ROC-AUC = 0.90, PR-AUC = 0.90). Among the Hybrid ALT400 ensembles, both top-3 and top-5 rank averaging reached accuracies of approximately 0.82 with ROC-AUC values of up to 0.90. All rank-based ensembles demonstrated better calibration and lower prediction variance than other fusion strategies.

These findings corroborate the fact that the integration of ML and DL models gives far superior generalization compared to individual model application. The hybrid ensembles captured both the feature-driven patterns of ML models and the high-dimensional nonlinear relationships of DL architectures, hence being more reliable for anticancer activity prediction using diverse descriptor sets.

Table 4.5. Comparative performance of ensemble machine learning and deep learning models on 2D and Hybrid datasets.

Data set	Feature selection	Ensemble	ML model	Accuracy	Recall	Precision	F1-macro	ROC-AUC	PR-AUC
2D	ALT400	rank_dlml_all	IDMLP-FTT-TabNet-XGB-ET-LGBM	0.81	0.81	0.81	0.81	0.90	0.90
		rank_dlml_top5	LGBM-TabNet-XGB-ET-IDMLP	0.81	0.81	0.81	0.81	0.90	0.89
		rank_dlml_top3	LGBM-TabNet-XGB	0.81	0.81	0.81	0.81	0.89	0.89
		median_dlml_all	IDMLP-FTT-TabNet	0.79	0.79	0.82	0.79	0.90	0.89
		soft_dlml_all	IDMLP-FTT	0.79	0.79	0.81	0.78	0.90	0.89

			TabNet-XGB-ET-LGBM						
		soft_vote_weighted_dlml_all	IDMLP-FTT-, TabNet-XGB-ET-LGBM	0.78	0.78	0.81	0.78	0.89	0.89
		stack_dlml_all	IDMLP-FTT-TabNet-XGB-ET-LGBM	0.78	0.78	0.81	0.77	0.89	0.89
2D	**MI400**	**rank_dlml_top3**	**XGB-TabNet-IDMLP**	**0.83**	**0.83**	**0.83**	**0.83**	**0.90**	**0.89**
		rank_dlml_all	IDMLP-FTT-TabNet-XGB	0.81	0.81	0.81	0.81	0.89	0.88

4.3. Final Model Selection

Final model selection was done based on both the quantitative performance and interpretability across all the ML, DL, and hybrid ensemble configurations. The models were evaluated on the ROC-AUC, PR-AUC, F1-macro, accuracy, and recall, with an emphasis on recall to minimize false negatives in anticancer prediction. Several ensembles demonstrated strong generalization, achieving an accuracy and recall greater than 0.80, and ROC-AUC values near 0.89. The final model chosen was the Rank-DLML-Top3 2D MI400 ensemble, since it demonstrated almost consistent superior performance with much lower variability across folds (Accuracy = 0.82, Recall = 0.82, F1-macro = 0.82, ROC-AUC = 0.89, PR-AUC = 0.89). This ensemble successfully combined the strongest ML model, (XGBoost), with top-performing DL architectures, (Improved Deep MLP and TabNet), ensuring robust performance with high recall**.** The selected models were subsequently used for external validation and explainability analyses, including SHAP value computation and global feature importance visualization.

4.4. External validation

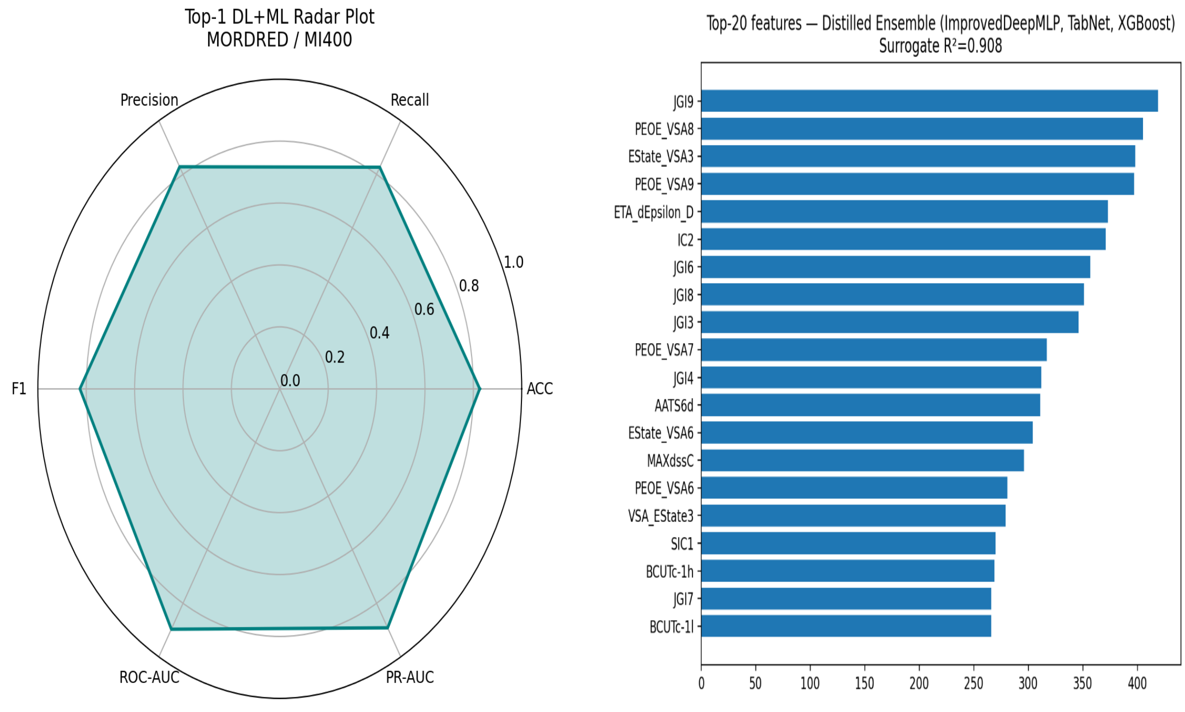
The final selected model was used for external validation to assess its ability to generalize to unseen chemical structures[38]. An external set of FDA-approved anticancer compounds and non-anticancer compounds was used to evaluate the model's real-world predictive reliability. The validation set consisted of 10 molecules (5 FDA-approved anticancer compounds and 5 non-anticancer compounds). As summarized in Table 4.6**,** the model correctly identified all five anticancer compounds (Erdafitinib, Tovorafenib, Abemaciclib, Tucatinib, and Encorafenib), achieving 100% recall for the active class. Among the non-anticancer compounds, three were correctly predicted as inactive, while two (Meclofenamate and Celecoxib) were misclassified as active, resulting in an overall accuracy of 80%. These results demonstrate that the final ensemble model generalizes effectively to FDA-approved compounds and maintains strong recall- an essential feature for anticancer screening, where minimizing false negatives is more critical than occasional false positives [39].

Table 4.6. External Validation result of 10 FDA drugs using the top-performing ensemble model (Improved Deep-MLP, TabNet, XGBoost -2D-MI400)

Compound Name	Actual Value	Probability of Ensemble Model	Predicted Value
Erdafitinib	1	0.65	1
Tovorafenib	1	0.60	1
Abemaciclib	1	0.82	1
Tucatinib	1	1.0	1
Encorafenib	1	0.56	1
Ketoprofen	0	0.43	0

Meclofenamate	0	0.56	1
Fenoprofen	0	0	0
Indomethacin	0	0.34	0
Celecoxib	0	0.86	1

4.5. Feature analysis and Model Interpretability



(a) (b)

Figure 4.7. Performance and interpretability analysis of the top-performing ensemble model.

To further interpret the behaviour of the final ensemble model, a performance summary and feature attribution analysis are conducted, as depicted in Figure 4.7. Figure 8a presents the radar plot that underlines the top-performing ensemble with very strong and well-rounded performance on the 2D MI400 feature set (Improved Deep MLP + TabNet + XGBoost), as all metrics, from accuracy and recall to precision, F1-macro, ROC-AUC, and PR-AUC, all show uniformly high values. This suggests that the presented ensemble achieves not only high predictive accuracy but also maintains uniform performance across all the measures. Figure 8b represents the results of the SHAP-based feature analysis, ranking the top 20 molecular descriptors contributing most to the predictions by the ensemble. Top-ranked descriptors include topological indices such as (JGI9, JGI6, and JGI8) electronic and charge-related features like (PEOE_VSA8, PEOE_VSA9, and ETA_epsilon_D) and fragment-based VSA features such as (EState_VSA3 and VSA_EState3). This would thus indicate that both electronic distribution and topological complexity are important for defining anticancer activity. The strong surrogate model fit, $R^2 = 0.908$, further suggests that SHAP explanations are coherent with the decision patterns of the ensemble. Overall, the explainability analysis confirms that the hybrid ML-DL ensemble captures chemically meaningful patterns and is dependent on interpretable molecular descriptors relevant to anticancer activity [40][41].

4.6 Diabetic Retinopathy Detection Results

The system's retina disease detection module was tested with a hybrid deep learning and machine learning-based approach. The approach combines convolutional neural networks for feature learning with various machine learning classifiers for classification, allowing for effective detection of various levels of diabetic retinopathy. The effectiveness of the system was evaluated with three classes: No Diabetic Retinopathy (NDR), Moderate Diabetic Retinopathy (MDR) and Proliferative Diabetic Retinopathy (PDR), as shown in Table 4.7. The classification results overall indicate that the proposed CNN-based hybrid system has a high accuracy, especially in recognising normal and abnormal retinal images.

The Radial Basis Function (RBF) classifier recorded the highest accuracy for the NDR class (98.66%), showing a high level of performance in distinguishing healthy retinas. But it showed lower accuracy for moderate disease levels (62.65% for MDR, 82.44% for PDR), which implies lower generalization for complex disease conditions. The Support Vector Machine (SVM) classifier showed the best overall performance with accuracies of 96.66% (NDR), 81.64% (MDR) and 90.07% (PDR). This suggests its ability to effectively handle normal and diseased retinal images, making it a good predictor for clinical applications.

Another effective classifier was the Random Forest (RF), with accuracies of 96.66% (NDR), 76.58% (MDR), and 83.96% (PDR). Although its accuracy was slightly lower than SVM for moderate and advanced stages, it produced consistent results due to ensemble learning. On the other hand, the Naïve Bayes (NB) classifier showed relatively weaker results with 94.0% (NDR), 74.68% (MDR), and 71.75% (PDR). While fast, its probabilistic nature restricts its ability to learn complex patterns in retinal images.

In conclusion, the findings suggest that although deep learning models effectively capture retinal features, the selection of classifier plays a crucial role in prediction accuracy.

The CNN + SVM combination provides optimal performance and generalisation, especially for moderate and severe diabetic retinopathy stages. These results demonstrate the benefits of combining deep learning with traditional machine learning approaches to enable accurate and efficient diagnosis of retinal diseases in the system.

Table 4.7. Performance Evaluation of CNN-Based Diabetic Retinopathy Detection Using Different Classifiers

Classifier	Metric	NDR (%)	MDR (%)	PDR (%)	Weighted Average (%)
RF	Accuracy	96.66	76.58	83.96	85.64
RF	Precision	92.40	84.00	79.70	85.60
RF	Recall	96.70	76.60	84.00	85.60
RF	F-Measure	94.50	80.10	81.80	85.50
SVM	Accuracy	96.66	81.64	90.07	89.29
SVM	Precision	96.70	87.80	83.10	89.40
SVM	Recall	96.70	81.60	90.10	89.30
SVM	F-Measure	96.70	84.60	84.60	89.30
RBF	Accuracy	98.66	62.65	82.44	80.86
RBF	Precision	93.70	81.10	67.90	81.50
RBF	Recall	98.70	62.70	82.40	80.90
RBF	F-Measure	96.10	70.70	74.50	80.50
NB	Accuracy	94.00	74.68	71.75	80.40
NB	Precision	89.80	74.70	75.80	80.20
NB	Recall	94.00	74.70	71.80	80.40
NB	F-Measure	91.90	74.70	73.70	80.30

4.7 Web-based System Development and Results

4.7.1 Interface and Integration

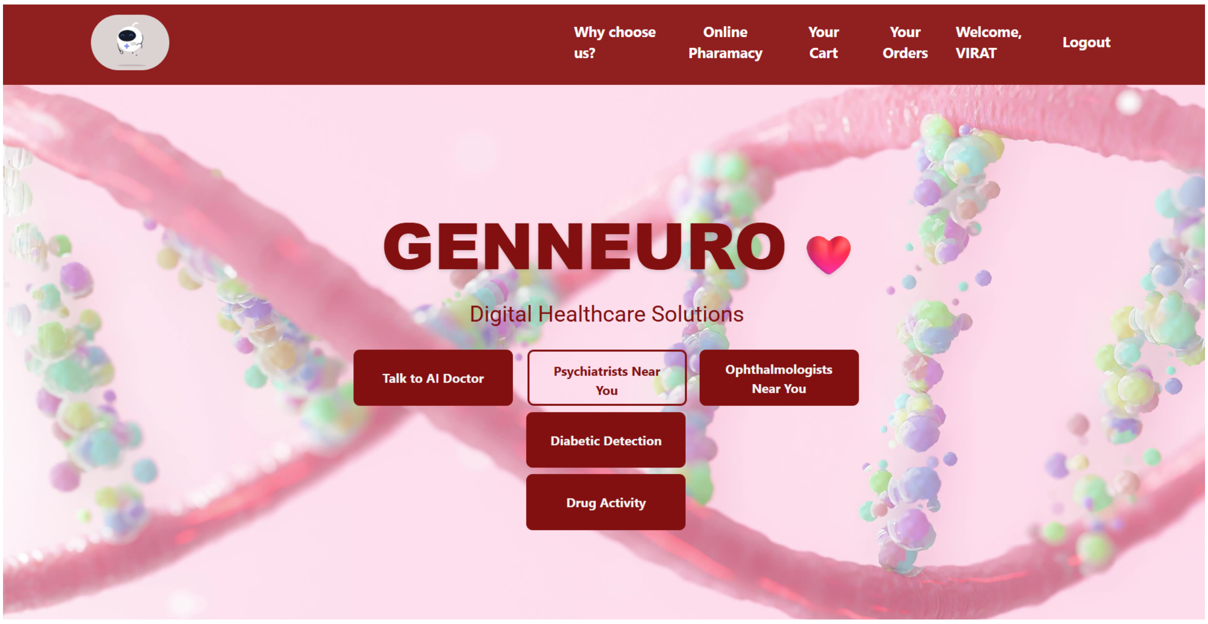


Figure 4.8: Web Interface

The system has been successfully implemented as a web-based integrated system that integrates several healthcare and AI-based modules. The dashboard serves as an entry point to services like AI consultation, detecting diabetic retinopathy, and predicting anti-cancer compounds. The user interface is user-friendly and responsive, catering to a diverse range of users with different levels of technical expertise. The single point of access to various services showcases the modular and scalable architecture of the system, enabling communication between front-end components and back-end AI services.

4.7.2 AI Chatbot and Mental Health

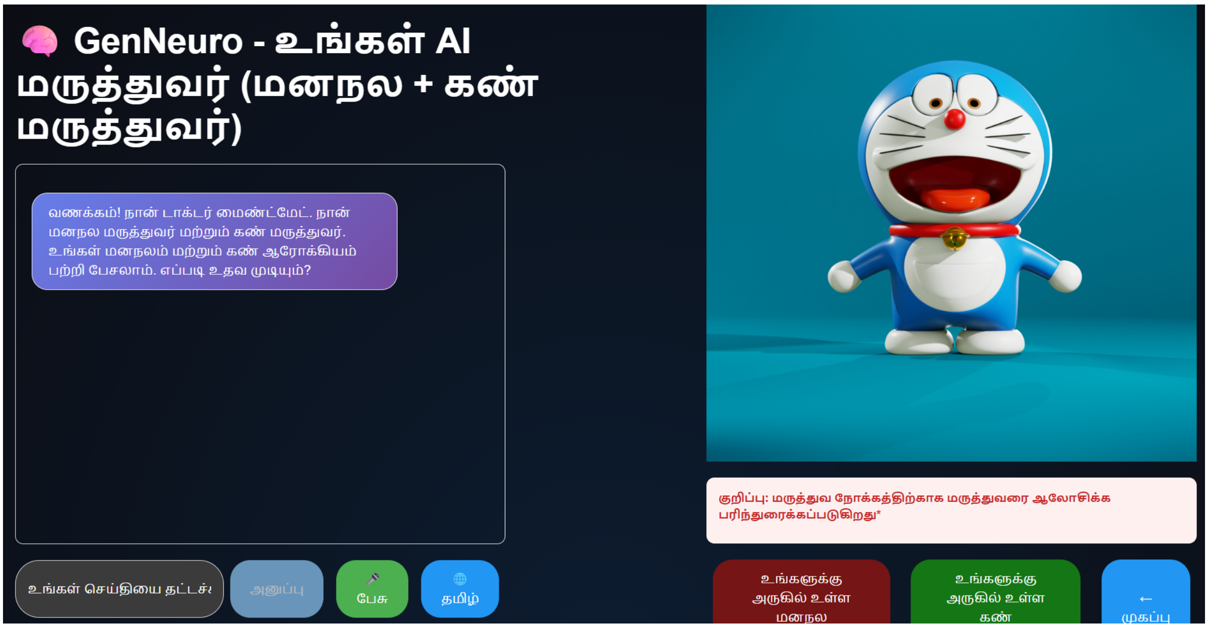


Figure 4.9: Bilingual Chatbot

The chatbot module offers improved user engagement through real-time communication and therapeutic support. The chatbot can comprehend user queries and respond accordingly, facilitating human-computer interaction. It enables multi-lingual conversations, thereby extending the system's reach. This module is a key component in understanding user interactions and emotions that help to provide initial mental health advice and enhance user engagement with the system.

4.7.3 Doctor and Specialist Recommendation Module

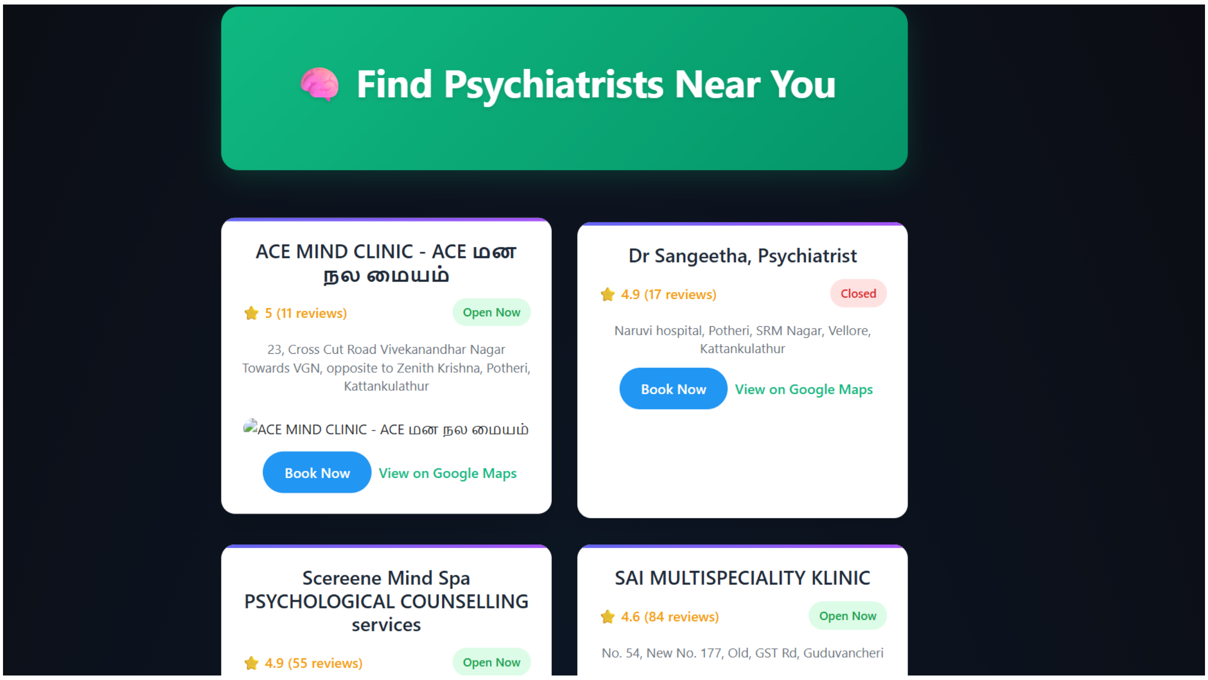


Figure 4.10: Psychiatrist Recommendation Module

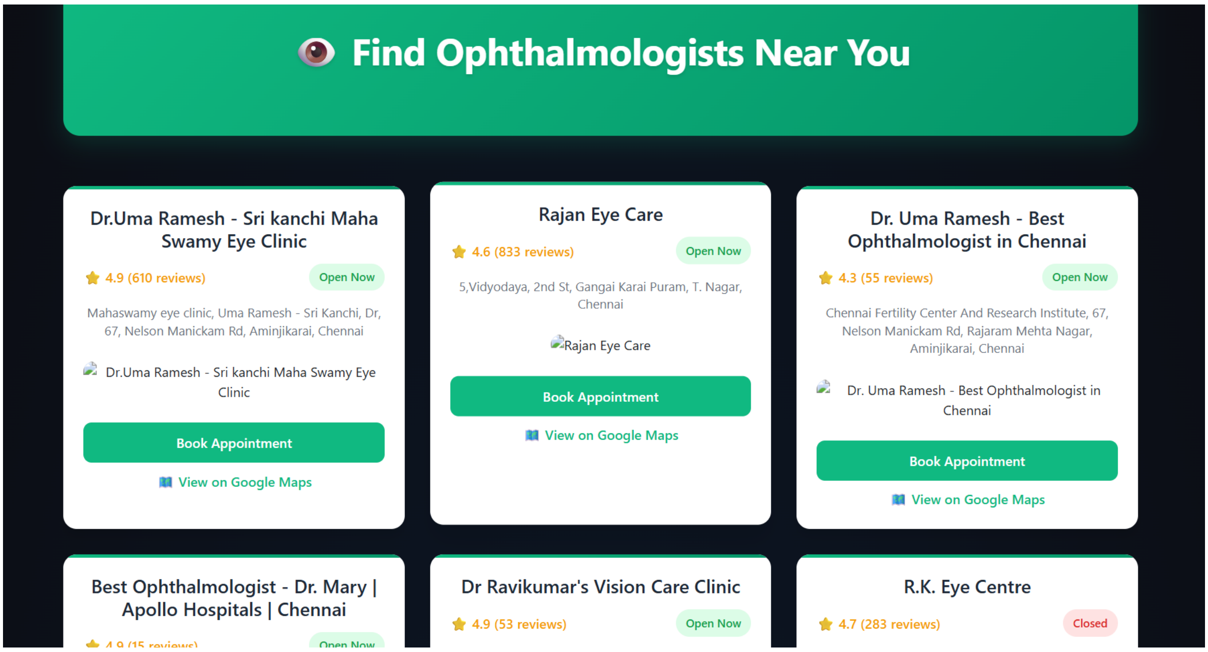


Figure 4.11: Ophthalmologists Recommendation Module

The system includes a recommendation module that provides users with access to local psychiatrists and ophthalmologists around 60km radius. Through the integration of external services like map services, it can provide up-to-date data on the availability, ratings and locations of doctors. This allows users to directly connect with healthcare providers based on their needs, thus connecting AI-driven predictions with medical help. The addition of scheduling and navigation features also adds to the system's convenience.

4.7.4 Online Pharmacy and E-Commerce Module

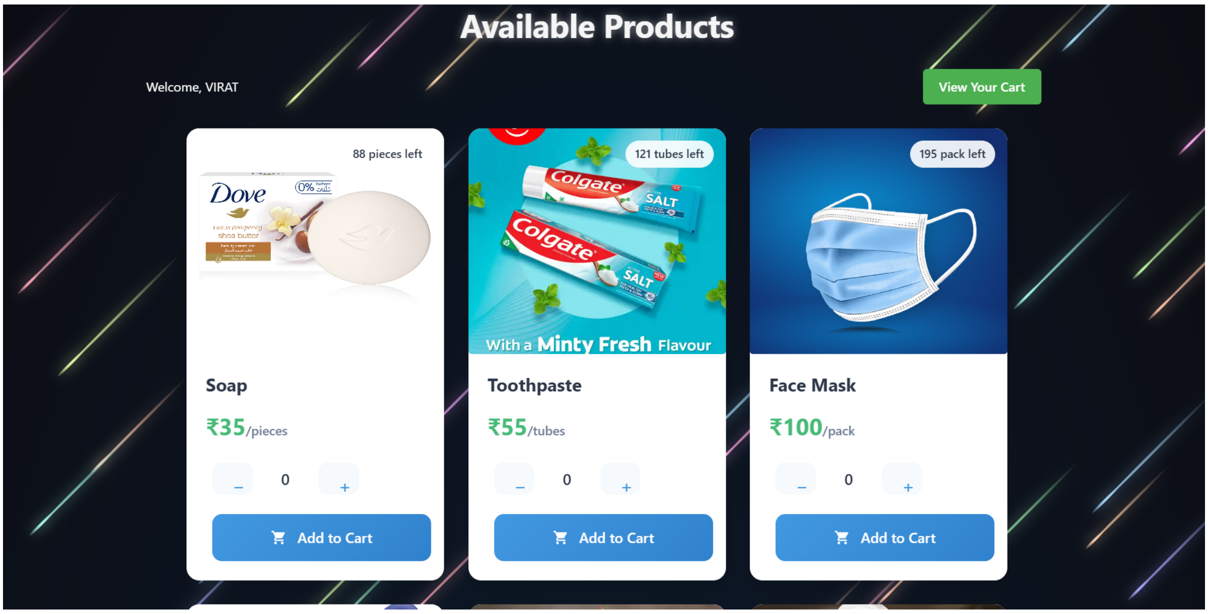


Figure 4.12: Online E-Commerce Module

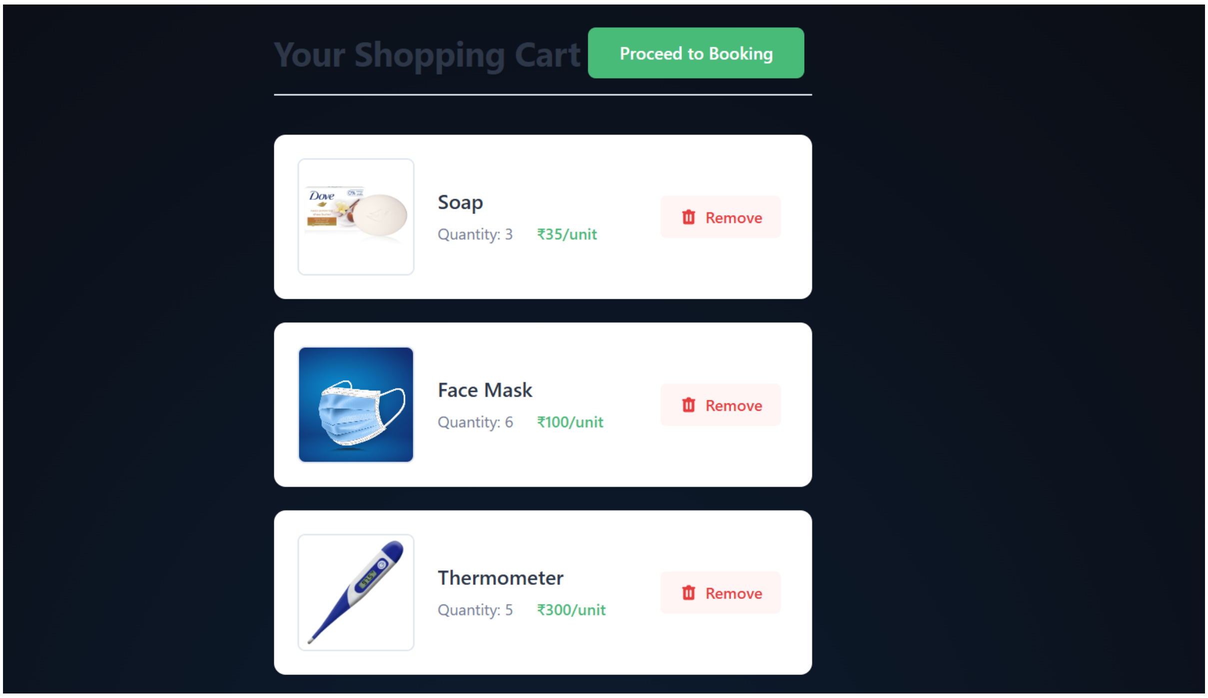


Figure 4.13: Online Pharmacy Module

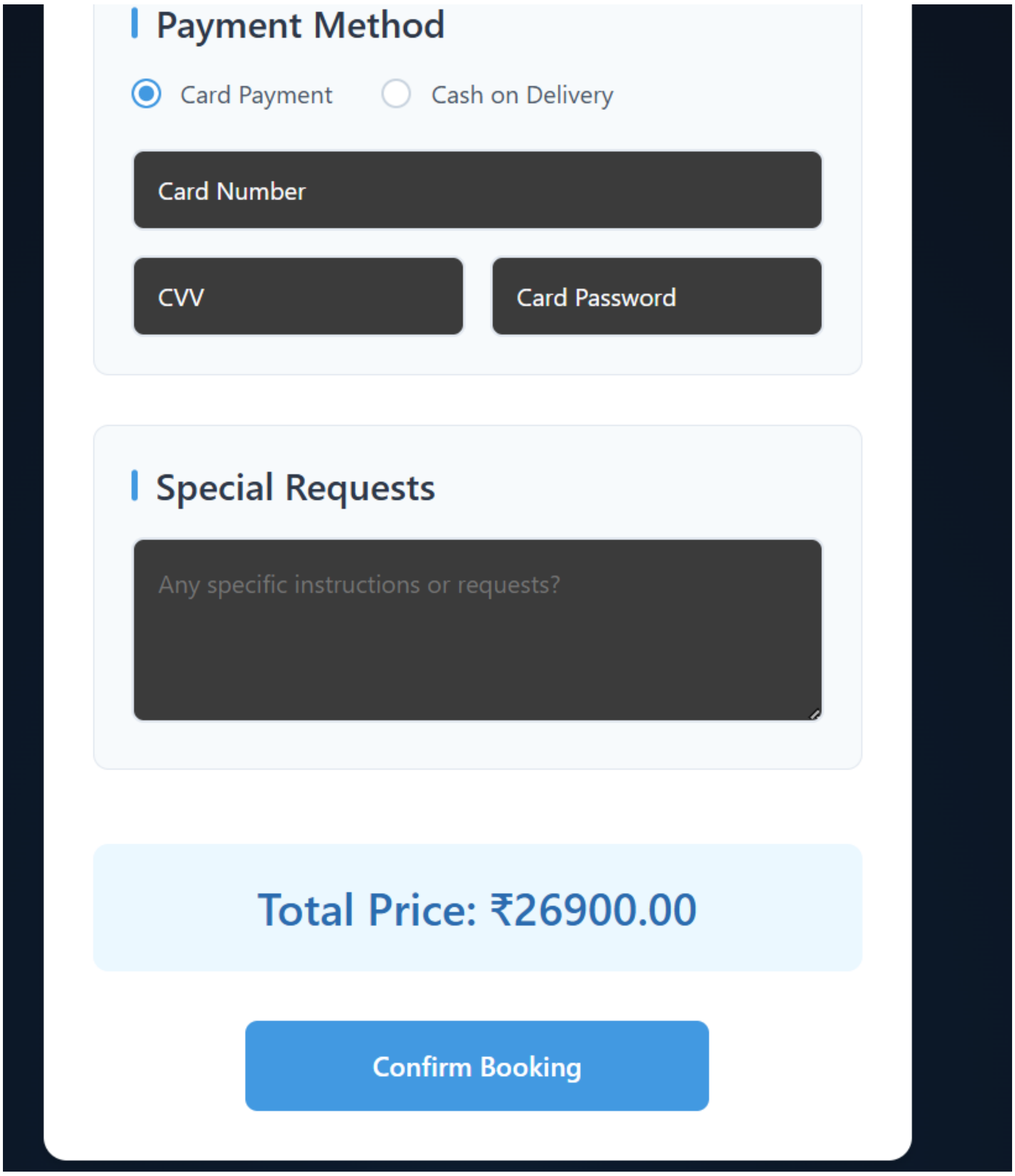


Figure 4.14: Online E-Commerce Payment Interface

A pharmacy module was designed to provide additional services in the delivery of medical care. This module enables users to search for medical products, select items and proceed with a purchase using a checkout process. Order tracking and transaction management enable an efficient workflow. This module showcases the system's potential to facilitate end-to-end medical services, leveraging AI insights and practical access to medical information.

4.7.5 CNN-Based Retinal Image Detection

Figure 4.15: Retinal Image Detection Web Interface

The retinal disease detection component allows users to upload the fundus image and retrieve the diagnosis for diabetic retinopathy and macular edema. The retinal image is fed into a convolutional neural network model, which produces the original and preprocessed images and the model's prediction. These show whether diseases are present or not, supplying users with clear diagnostic information. The model's good performance, as explained in the preceding sections, allows accurate real-time prediction.

4.7.6 Anti-Cancer Prediction Result

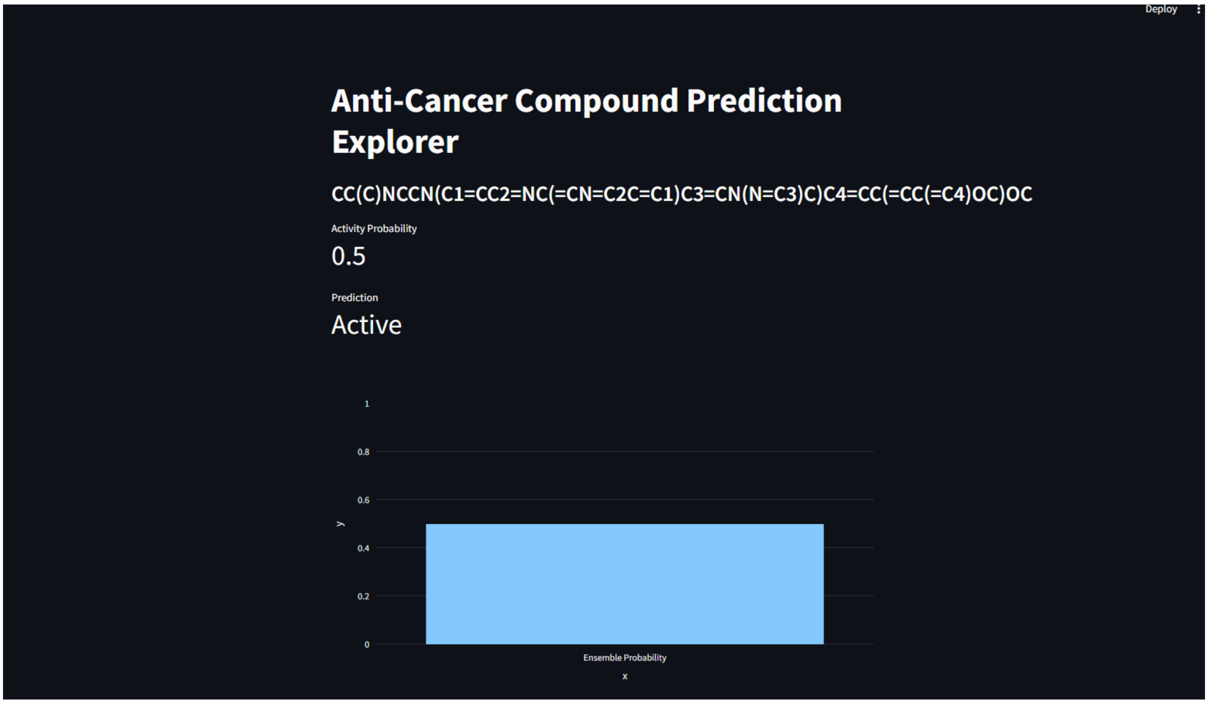


Figure 4.16: Anti-cancer Prediction Interface

The anti-cancer prediction model offers a user interface for assessing molecule activity with the built ML-DL ensemble model. Molecular inputs are processed and predictions are returned regarding activity (active vs. inactive) and probability values. The outcomes are displayed in an easy-to-understand manner and through visual representations of prediction confidence. This shows how the prediction models can be applied, and their potential use in drug discovery.

4.7.7 System-Level Discussion

The combination of several AI modules in the system highlights the benefits of a multimodal approach to health care and biomedical applications. The integration of conversational AI, image recognition, prediction, and service management offers a holistic approach that goes beyond the limitations of single-task models. The findings show that the system is technically sound and practically usable, providing timely and actionable insights and enhancing access to health services. This integrated approach underscores the benefits of combining machine and deep learning models with human-centred design, to solve complex problems in healthcare.

CHAPTER 5
CONCLUSION & FUTURE ENHANCEMENTS

Conclusion and Future Enhancements

The project created an artificial intelligence system which uses advanced machine learning and deep learning techniques to forecast anticancer small molecules. The system achieves precise evaluation of chemical compounds and their biological effects through its ability to process molecular descriptor data together with multiple methods of feature selection.

The application of feature engineering methods through Mutual Information (MI-400) and ALT-400 and PCA-400 made it possible to choose important features that would support model development. The team used various machine learning models to evaluate their performance until they found the most effective one which could accurately recall results required for biomedical research. The research team developed deep learning models which contained Improved Deep Multilayer Perceptron (MLP) and FT-Transformer and TabNet to identify complex nonlinear patterns within the dataset.

The Optuna framework used its hyperparameter optimization process to improve performance which enabled the model to handle multiple datasets. The model achieved better durability and improved evaluation techniques through its implementation of multiple training datasets together with various feature tracks.

The system will benefit from future enhancements which will enable it to use bigger datasets that contain more diverse information to create better generalization patterns and more trustworthy results. The system can achieve better molecular representation through advanced deep learning methods which include graph neural networks and these methods will improve prediction accuracy. The system can expand its capabilities through additional biomedical data sources which include genomic data and clinical data to deliver complete analysis solutions.

The healthcare platform requires improved user interface design together with real-time prediction functions and cloud-based deployment options to enhance system usability and capacity expansion. The system should use AutoML techniques together with advanced ensemble methods to achieve optimal performance results while decreasing the requirement for human intervention.

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APPENDIX
A. CODING

```
import os
import numpy as np
import pandas as pd

from sklearn.impute import SimpleImputer
from sklearn.feature_selection import VarianceThreshold, SelectKBest, mutual_info_classif
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.model_selection import StratifiedKFold
from sklearn.metrics import accuracy_score, f1_score, precision_score, recall_score


MORDRED_TRAIN = "dataset/ic50_train_mordred.csv"
N_KEEP = 400
MISSING_DROP_THR = 0.50


def detect_target(df):
    for c in df.columns:
        if c.lower() in ["cls","class","label","active","activity","y","target"]:
            return c
    return df.columns[-1]


def clean_numeric(df, target_col):
    missing = df.isna().mean()
    drop_missing = set(missing[missing > MISSING_DROP_THR].index)
```

```
    const = set(df.drop(columns=[target_col]).nunique(dropna=False)[lambda s: s <= 1].index)

    id_like = {c for c in df.columns if c.lower() in
["name","id","mol_id","molecule","smiles"]}

    to_drop = (drop_missing | const | id_like) - {target_col}
    keep = [c for c in df.columns if c not in to_drop and c != target_col]

    X = df[keep].apply(pd.to_numeric, errors="coerce")
    return X, keep


def preprocess_dataset(train_path):
    df = pd.read_csv(train_path)

    target = detect_target(df)
    y = df[target].astype(int)

    X_raw, kept_cols = clean_numeric(df, target)

    imp = SimpleImputer(strategy="median")
    X_imp = pd.DataFrame(imp.fit_transform(X_raw), columns=kept_cols)

    vt = VarianceThreshold(0.0)
    X_vt = vt.fit_transform(X_imp)
    vt_cols = np.array(kept_cols)[vt.get_support()]
    X_vt_df = pd.DataFrame(X_vt, columns=vt_cols)

    kA = min(N_KEEP, X_vt_df.shape[1])
    sel = SelectKBest(mutual_info_classif, k=kA)
    X_mi = sel.fit_transform(X_vt_df, y)

    l1 = LogisticRegression(penalty="l1", solver="saga", max_iter=5000)
    l1.fit(X_vt_df, y)
```

```
    coef_mask = np.abs(l1.coef_).ravel() > 1e-8

    if not np.any(coef_mask):
        lasso_cols = vt_cols
    else:
        lasso_cols = vt_cols[coef_mask]

    X_l1 = X_vt_df[lasso_cols]

    rf = RandomForestClassifier(n_estimators=400, random_state=42)
    rf.fit(X_l1, y)

    importances = pd.Series(rf.feature_importances_, index=lasso_cols)
    top_features = importances.sort_values(ascending=False).head(min(N_KEEP,
len(importances)))

    X_alt = X_l1[top_features.index]

    scaler = StandardScaler()
    X_std = scaler.fit_transform(X_imp)

    pca = PCA(n_components=min(N_KEEP, X_std.shape[1]), random_state=42)
    X_pca = pca.fit_transform(X_std)

    return pd.DataFrame(X_mi), X_alt, pd.DataFrame(X_pca), y


def train_model(X, y):
    skf = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

    results = []
    model = RandomForestClassifier(n_estimators=200, random_state=42)
```

```
    for train_idx, val_idx in skf.split(X, y):
        X_train, X_val = X.iloc[train_idx], X.iloc[val_idx]
        y_train, y_val = y.iloc[train_idx], y.iloc[val_idx]

        model.fit(X_train, y_train)
        y_pred = model.predict(X_val)

        acc = accuracy_score(y_val, y_pred)
        f1 = f1_score(y_val, y_pred, average="macro")
        precision = precision_score(y_val, y_pred, average="macro")
        recall = recall_score(y_val, y_pred, average="macro")

        results.append([acc, f1, precision, recall])

    return np.mean(results, axis=0)


def main():
    print("Processing dataset...")

    X_mi, X_alt, X_pca, y = preprocess_dataset(MORDRED_TRAIN)

    print("Training model...")

    acc, f1, prec, rec = train_model(X_mi, y)

    print("\nFinal Results:")
    print("Accuracy:", acc)
    print("F1 Score:", f1)
    print("Precision:", prec)
    print("Recall:", rec)


if __name__ == "__main__":
```

B. CONFERENCE PUBLICATION

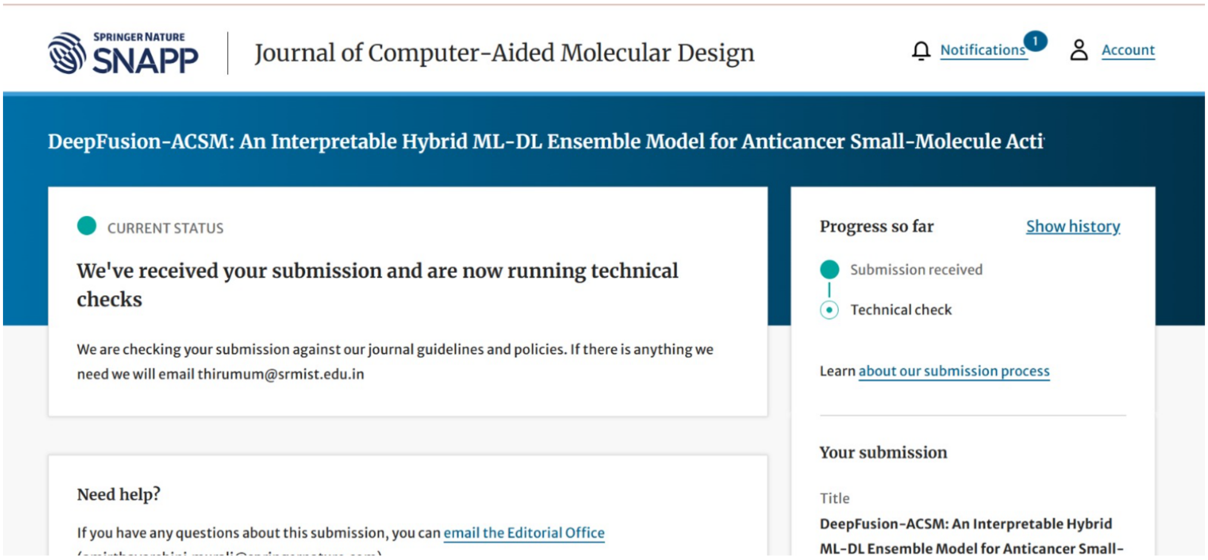

Figure B.1: Journal Submission

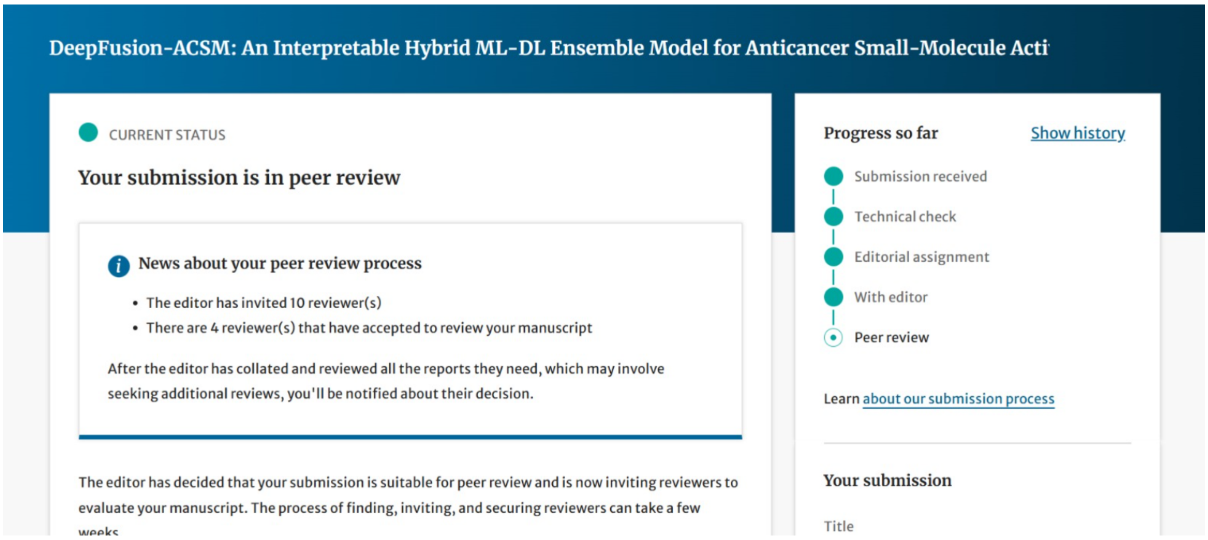

Figure B.2: Journal Under Peer Review

C. JOURNAL PUBLICATION

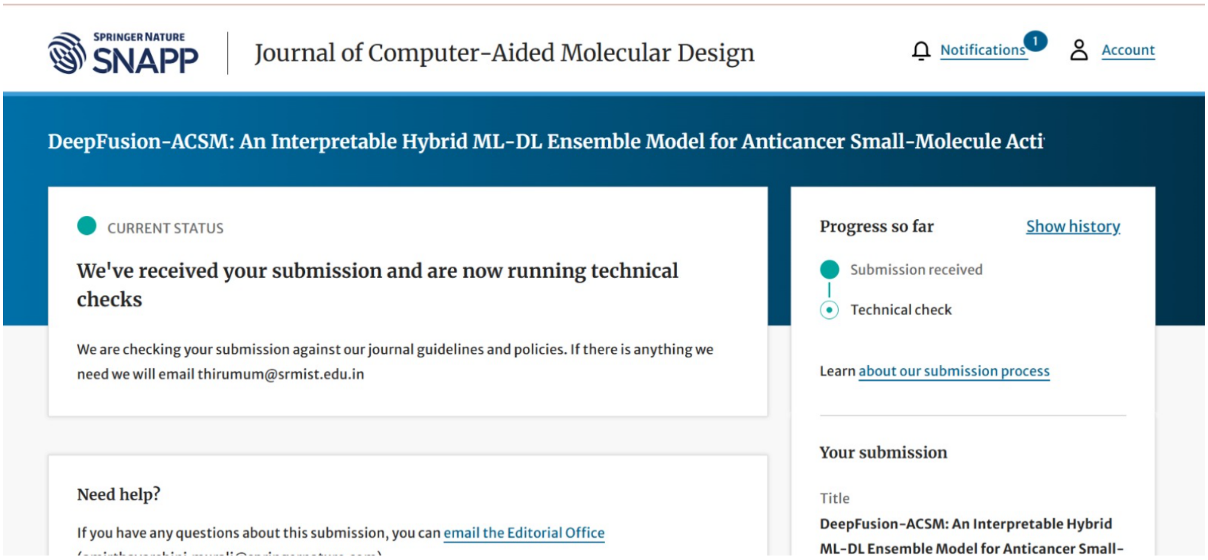

Figure C.1: Journal Submission

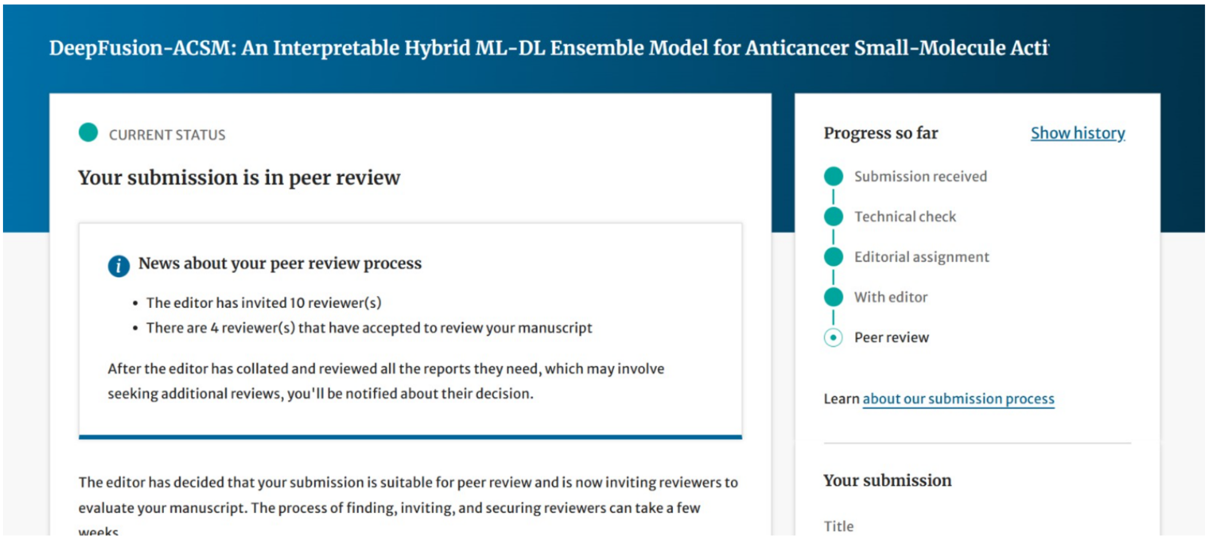

Figure C.2: Journal Under Peer Review

D. PLAGIARISM REPORT

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